

DHF Example Portfolio

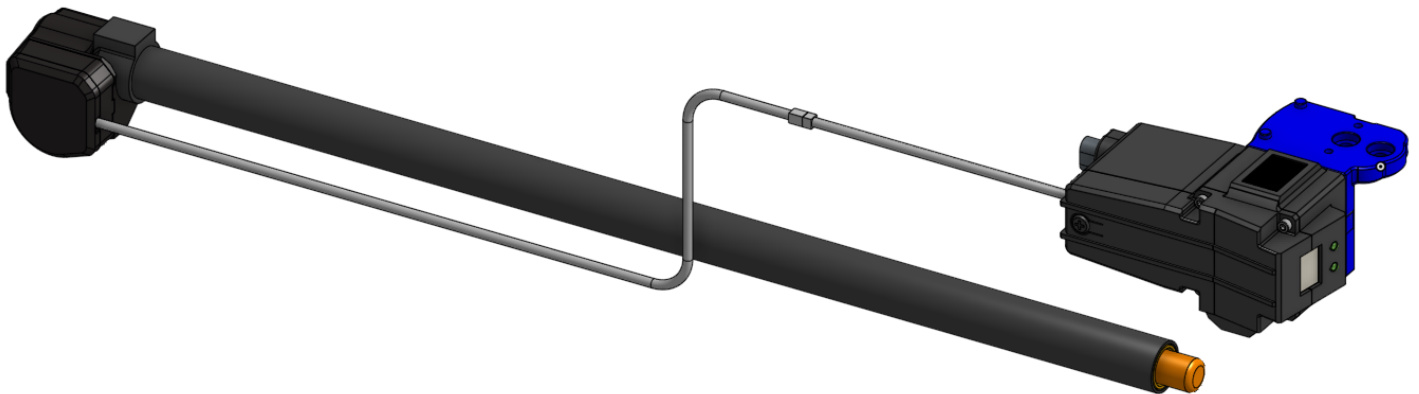
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Ammunition Storage Display Unit

HMX SDU1



PORTFOLIO READERS GUIDE AND DHF INDEX HMX SDU1

DHF-[FAKEDHF#]

This portfolio contains a **Design History File (DHF)-style** document set for the **HMX SDU1**, an ammo storage display unit for the Henry Model X firearm. The HMX SDU1 is **not a medical device**; it is a consumer firearm accessory sold through my small business Sovereigness Designs. The intent of this package is to demonstrate how I execute medical device design controls in practice: planning, requirements, risk management, traceability, design, verification/validation, and design transfer - while scaling the rigor to match a non-regulated, low-volume product made by an owner-operator single employee business.

This is an abbreviated packet showing only the items from section 4. The full DHF or native files are available upon request.

1. What this is intended to demonstrate

- **Fluency with design controls:** I can step into an established QMS and operate within existing procedures with minimal ramp-up.
- **Strong engineering execution:** Design outputs (drawings/specs) and test evidence reflect real engineering work and familiarity with the full design cycle.
- **Clear technical writing:** Documents are written to be reviewable and auditable in style, even where simplified.
- **Self-directed project leadership:** I can define scope, drive deliverables, and close the loop from requirements to evidence.

2. What this is (and is not)

- **This is not a claim of regulatory compliance** for the HMX SDU1.
- **This is not an FDA submission** and is not prepared under a certified QMS.
- The document structure and intent **mimic** the content and flow expected in regulated medical device development to showcase competency. Some humor is present.

3. How this package is scoped (intentional simplifications):

- **Sample sizes are small** and justified by low-volume manufacturing and low-risk use environment. Conventionally sample size, TMV/MSA, and confidence levels would be risk-based and formally planned.
- **Supplier controls and process validations are streamlined.** Where applicable, controls are represented through drawings, inspection points, and work instructions rather than full supplier qualification and PV protocols.
- **Traceability and Data Verification are minimal.** My small business does not serialize for traceability of all product as would be appropriate for a medical device. Equipment is not actually calibrated by a metrologist. Component lots are not carefully managed.

4. How to read this quickly - If you only have 15 minutes, the best “story of the product” is:

1. **Design & Development Plan SD1001**
2. **Trace Matrix SD1002**
3. **Design Outputs DO- documents**
4. **Design Verification Report SD1009**

DHF Index		
Document #	Document Title	Revision
SD1001	Design and Development Plan	00
SD1002	Trace Matrix	02
SD1003	Design FMEA	00
SD1004	Usability FMEA	00
SD1005	Device Master Record	00
SD1006	Design Verification and Validation Plan	00
SD1007	HMX SDU1 Process Flow Diagram	00
SD1008	Design Verification Protocol	00
SD1009	Design Verification Report	00
SD1010	Design Validation Protocol	00
SD1011	Design Validation Report	00
SD1012	Process FMEA	00
SD1014	Risk Assessment	00
SD1015	Design Transfer Checklist	00
SD1016	Work Instruction Manufacture HMX SDU1	00
DO1001	HMX SDU1 Top Level Assembly	A
DO1002	Bill Of Materials	B
DO1003	HMX SDU1 Software	A
DO1004	Circuit Diagram	A
DO1006	DISPLAY UNIT	A
DO1007	SENSOR MAGAZINE	B
DO1008	RMR OPTIC MOUNT	A
DO1009	HOUSING	A
DO1010	HOUSING COVER	A
DO1011	CONTROLLER	A
DO1012	RESISTOR BOARD	A
DO1013	SENSOR CONNECTOR	A
DO1014	SPRING SPACER	A
DO1015	MODIFIED STOCK MAGAZINE END CAP	A
DO1016	FOLLOWER CABLE CONNECTOR	A
DO1017	TUBE MOUNT	A
DO1018	SPOOL CASE	A
DO1019	SPOOL CASE COVER	A
DO1020	ENCODER SHAFT	A
DO1021	SENSOR UNIT COVER	A
DO1022	ENCODER AND CONNECTOR	A
DO1023	PRINTED PART GCODE	A
DO1024	INSTRUCTIONS FOR USE HMX SDU1	00

DESIGN AND DEVELOPMENT PLAN - AMMUNITION STORAGE DISPLAY UNIT FOR LEVER ACTION RIFLE

DHF-[FAKEDHF#]

1. PURPOSE

The purpose of this document is to plan the design and development of an Ammunition Storage Display Unit (Ammo SDU) for use in lever action rifles or other tube-magazine fed firearms.

2. PROJECT OBJECTIVES

Lever-action rifles and other tube magazine fed firearms store ammunition in a fixed, opaque linear tube typically mounted parallel to the barrel on the underside of the firearm. It is easy to check or clear the chamber at a glance but it is typically not possible to determine the number of rounds in storage without fully unloading the firearm. Knowing how many rounds remain in the magazine at any given time is useful information for recreational or competitive shooting.

The objective of this project is to develop a device to detect and display to the user the number of rounds stored in the magazine tube, in a manner readily accessible to the user in typical shooting activities, without otherwise altering the functioning of the firearm.

3. PROJECT SCOPE

The project scope includes:

- A self-contained (removable) unit for detecting and displaying the number of rounds in the tube magazine of a Henry Model X .357 Rifle. (Part number: HMX SDU1)
- Manufacturing processes appropriate for custom-to-order volume up to 2 units/month.
- An open-source repository for software.

Compatibility with other rifles or magazines may be added under a separate project.

Manufacturing processes will scale under separate projects as needed.

4. PROJECT ROLES

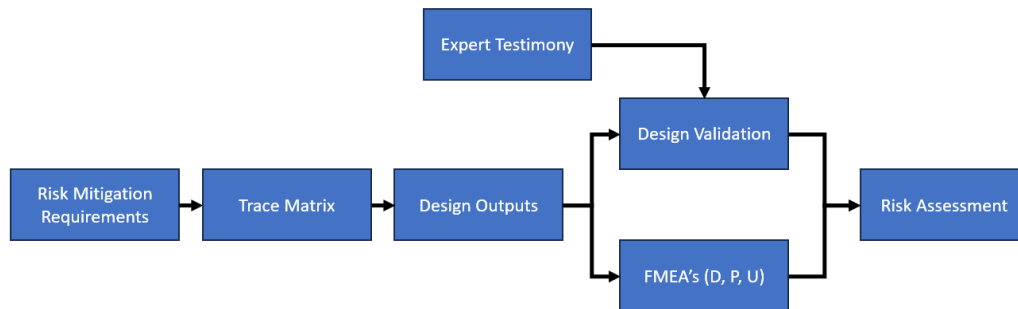
Role	Person
Project Manager	Jessica Boyers
Marketing	Jess Boyers
R&D	J. Boyers
Manufacturing	J M Boyers
Design Assurance	Boyers, Jessica

5. PROJECT PLANNING

5.1 Quality Planning

Sovereigness Designs does not operate under a Quality Management System. Standard Good Documentation Practices shall be used for document and records creation.

Risk management in this case shall be driven by the FMEA's culminating in the Risk Assessment. Risk mitigation requirements shall be included in the Trace Matrix. Design Validation shall include assessment of the risk acceptability from qualified experts under simulated use.



[Author's aside: This is not representative of a typical medical device Risk management procedure due to the nature of the device]

5.2 Manufacturing Planning

This device is produced and sold custom-to-order, and will not be mass manufactured. Any Fit and Finish requirements shall be captured in the Trace Matrix. Manufacturing techniques and requirements particular to each part shall be captured in the DMR.

5.3 Clinical Planning

This device should not be used in a clinical setting excepting cases of alien incursion or zombification, so no clinical planning shall be performed. Range Safety Officers shall be consulted in the creation of Verification and Validation protocols that take place in Range conditions.

5.4 Regulatory Planning

This device is not a medical device. As a firearm accessory, in North Carolina there are no regulations pertaining to the design, sale, or use of firearm accessories (not including ammunition) that do not attach to the barrel or interact with the trigger or firing group.

5.5 Human Factors (HF) Planning

Requirements for User Interface shall be included in the Trace Matrix as applicable. The Usability FMEA based on the Design Outputs shall drive the identification of critical tasks with a risk-based approach. Critical task completion and user evaluation shall be included in the Design Validation activities.

5.6 Post-Market Surveillance Planning

In a medical device context, this would include strategies for tracking device effectiveness, complaints, and adverse events after launch including durations and information systems, among other things.

In this context, Customer Service and Support. Sovereigness Designs shall provide installation instructions, disassembly instructions, and after-purchase troubleshooting assistance to customers. SD shall also offer repair or replacement within 1 year of purchase for defects in manufacture. SD shall publish and maintain a free and open-source repository for the device software and any updates.

5.7 Software Development Planning

Software shall be developed in the Arduino language variant of C++. A.I. tools may be utilized in the creation of software but all outputs shall be human verified.

6. DELIVERABLES CHECKLIST

Phase	Required Deliverables	Document Reference	Comments/Rationale
Phase 1-2 – Plan, UNs, DIs	Design and Development Plan	SD1001	
	Risk Management Plan	SD1001	
	Trace Matrix (Initial)	SD1002	
	Design Review	N/A	Inapplicable in this context, but normally critical.
Phase 3 – Design Outputs	Drawings, Models, Specs, IFU	See DMR	
	Software	See DMR	
	Usability FMEA	SD1004	
	Design FMEA	SD1003	
	Device Master Record (Initial)	SD1005	
	Design Verification and Validation Plan	SD1006	
	Design Review	N/A	Inapplicable in this context, but normally critical.
Phase 4 – Design Verification	Design Verification Protocol	SD1008	
	Design Verification Report	SD1009	
	Trace Matrix (Update)	SD1002	
	Design Review	N/A	Combined with Design Validation Review
Phase 5 – Design Validation	Design Validation Protocol	SD1010	
	Design Validation Report	SD1011	
	Trace Matrix (Final)	SD1002	
	Design Review	N/A	Inapplicable in this context, but normally critical.
Phase 6 – Design Transfer	Process Flow Diagram	SD1007	
	Process FMEA	SD1012	
	Process Validation Plan	SD1012	
	Process Validation	N/A	Per rationale in SD1012, no process validation will take place.
	Work Instruction, Manufacture HMX SDU1	SD1016	
	Risk Assessment	SD1014	
	Device Master Record (Final)	SD1005	
	Design Transfer Checklist	SD1015	
	Design Review	N/A	Inapplicable in this context, but normally critical.

Phase	Required Deliverables	Document Reference	Comments/Rationale
Phase 7 – Post Release	Post-Market Surveillance Plan	SD1001	In this context, customer service and support.

7. ATTACHMENTS

- ☒ No Attachments If Checked

8. REVISION HISTORY

Revision	Revision Date	Revision Description
00	14-APR-2025	Initial Release

Traceability Matrix - General Information
Product Name: Ammunition Storage Display Unit (Ammo SDU) DHF/Index Number: DHF-[FAKEDHF#] Document Number: SD1002 Revision: 02

Scope:	HMX SDU1
Referenced Documents:	SD1001

Intended Use	The HMX SDU1 is a self-contained unit for detecting and displaying the number of rounds stored within the tube magazine of a Henry Model X .357. The HMX SDU1 maintains a real-time count, and updates the display as rounds are loaded or unloaded from the magazine tube. The HMX SDU1 can be assembled with a Henry Model X .357 by an armorer or any adult user with the provided assembly instructions. The HMX SDU1 is intended for use by any adult user of a Henry Model X .357 and in any environment appropriate to the use of a Henry Model X .357. The HMX SDU1 is intended for cosmetic, recreational, and competitive shooting purposes. It is not intended as a safety device and can never be used in lieu of standard firearm clearance practices. It is not intended as a self-defense or hunting aid and can not be relied on for ready-to-fire verification.
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Approvals

Functional Area	Printed Name	Signature	Date
R&D	Jessica Boyers		
Design Assurance	Jessica Boyers		
Manufacturing	Jessica Boyers		
Marketing	Jessica Boyers		

Revision History

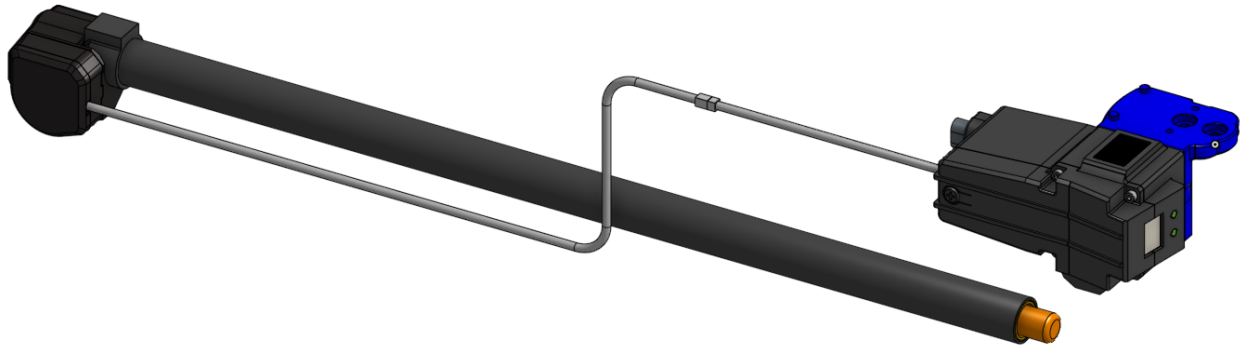
Revision	Revision Date	Revision Description
00	13-APR-2025	Initial Release
1	27-APR-2025	Added Design Outputs
2	27-OCT-2025	Added DV&V reports and final comments.

User Need ID	User Need Description	User Need Source	Design Input ID	Design Input Requirement	Design Input Rationale	Design Output	Verification Report Number	Validation Report Number	Comments
UN01	The user needs the device to be compatible with the Henry Model X .357.	VoC	DI-01-001	The device shall interface with the magazine well of the Henry Model X .357 in stock condition.	Should not require modification to the rifle to use device.	DO1001	SD1009	SD1010	N/A
			DI-01-002	The device shall operate with .357 and .38 special ammunition.	Henry Model X .357 can use both kinds of ammunition.	DO1001	SD1009	SD1010	N/A
			DI-01-003	The device shall mount to standard m-lok, picatinny, or JDR pattern mounts.	Henry Model X ships with mounting hardpoints for a number of accessory ecosystems and adapters are readily available.	DO1001, DO1008	SD1009	SD1010	N/A
UN02	The user needs the device to be removable.	VoC	DI-02-001	The magazine end of the device shall connect and disconnect with the firearm identically to the stock magazine.	Removing the device should be no harder than removing other common firearm accessories.	DO1001, DO1015, DO1017	SD1009	SD1010	N/A
			DI-02-002	The device shall mount to standard m-lok, picatinny, or JDR pattern mounts using removable fasteners.	So that the firearm can be returned to stock condition if desired.	DO1001, DO1008	SD1009	SD1010	N/A
UN03	The user needs the device to not alter or interfere with the normal functioning of the firearm (loading, unloading, cycling, firing, clearing)	Regulatory, Safety	DI-03-001	The device shall not obscure or prevent the use of an optic.	To allow normal aiming.	DO1001, DO1008	SD1009	SD1010	N/A
			DI-03-002	The device shall not obstruct the magazine follower during loading and unloading.	To allow normal cycling.	DO1007	SD1009	SD1010	N/A
			DI-03-003	With the device installed, 7 rounds shall be able to be stored in the magazine tube.	To maintain minimum ammunition capacity.	DO1001, DO1014	SD1009	SD1010	This is one round reduced from the standard capacity - this is an acceptable tradeoff for the device.
			DI-03-004	With the device installed, a cyclic fire rate of at most 3 seconds per round shall be achievable.	To maintain shooting pace of unmodified Henry Model X.	DO1001	SD1009	SD1010	N/A
			DI-03-005	The device shall not obstruct the bolt and ejection window.	To maintain normal shooting and shooting safety.	DO1001	SD1009	SD1010	N/A
			DI-03-006	The device shall not obstruct or attach to the barrel.	To maintain normal safety and be in regulatory compliance.	DO1007, DO1017	SD1009	SD1010	N/A
			DI-03-007	The device shall not attach to or modify the trigger or fire control group.	To be in regulatory compliance.	DO1001	SD1009	SD1010	N/A
			DI-03-008	The device shall not obstruct the charging lever.	To maintain normal loading.	DO1001	SD1009	SD1010	N/A
			DI-03-009	The device shall be powered by an on-board battery.	So no wires tether the firearm to external battery or source.	DO1002, DO1006	SD1009	SD1010	N/A
			DI-03-010	The device shall not obstruct or interfere with the trigger safety or lever-deactivator.	To maintain shooting safety.	DO1001	SD1009	SD1010	N/A
UN04	The user needs the device to display the current number of rounds in the magazine.	VoC	DI-04-001	The device shall be capable of displaying the numerals 0-7 visibly in daylight conditions.	Main device functionality.	DO1006	SD1009	SD1010	N/A
			DI-04-002	The device shall display 0 when the magazine is empty.	Shall not be used in place of normal clearance verification.	DO1003	SD1009	SD1010	N/A
			DI-04-003	The device shall increment each time a round is loaded into the magazine.	Main device functionality.	DO1003	SD1009	SD1010	N/A
			DI-04-004	The device shall decrement each time a round is removed from the magazine.	Main device functionality.	DO1003	SD1009	SD1010	N/A
			DI-04-005	The device shall have an indicator for round-in-battery.	As a shooting aid.	DO1003	SD1009	SD1010	This indicator does not directly check the chamber (it would be unsafe to breach the battery) but its status is inferred from magazine status changes.
			DI-04-006	The device shall have error handling for over-load, jolt, connection loss and count loss.	To account for foreseeable misuse and exceptional circumstances.	DO1003	SD1009	SD1010	N/A
			DI-04-007	The device shall have manual controls for resetting and maxing the count.	To correct a sensor-magazine de-sync.	DO1003, DO1006, DO1010	SD1009	SD1010	N/A
			DI-04-008	The device shall maintain the appropriate count when the firearm is fired.	Main device functionality.	DO1003	SD1009	SD1010	N/A
			DI-04-009	The device shall be operable continuously for at least 1 hour.	This includes battery life.	DO1006	SD1009	SD1010	N/A
UN05	The user needs the device to be appropriate for use in normal firing and handling positions.	VoC	DI-05-001	The display shall face the user when held in a firing position.	To be seen by the firearm user.	DO1006	SD1009	SD1010	N/A
			DI-05-002	The display shall be viewable with the firearm shouldered and standard sight picture acquired.	So ammo can be read without losing sighting.	DO1001	SD1009	SD1010	N/A
			DI-05-003	The display shall be mounted on the opposite side of the firearm as the ejection port.	To prevent ejected round collisions.	DO1006, DO1008	SD1009	SD1010	N/A
			DI-05-004	The device shall increase the weight of the firearm by less than or equal to 3 lbs.	To maintain usability ergonomics.	DO1001	SD1009	SD1010	N/A
			DI-05-005	The device controls shall be operable with a single hand.	So that the firearm does not need to be set down to operate.	DO1006	SD1009	SD1010	N/A
			DI-05-006	The device shall not obstruct the handle or fore-grip.	To maintain a normal grip.	DO1007	SD1009	SD1010	N/A
UN06	The user needs the device to withstand normal environmental exposures.	VoC	DI-06-001	The device shall allow limited dust ingress and exclude foreign objects.	For use in normal firearm conditions such as outdoor ranges.	DO1006, DO1007	SD1009	SD1010	IP44 is NOT claimed.
			DI-06-002	The device shall prevent entry of rain water.	For use in normal firearm conditions such as outdoor ranges.	DO1006, DO1007	SD1009	SD1010	IP44 is NOT claimed.

DMR SUMMARY HMX SDU1

DHF-[FAKEDHF#]

Purpose	This document summarizes the contents of the Device Master Record for product SKU HMX SDU1. This document and all referenced items comprise the DMR.
Scope	This document applies to SKU HMX SDU1.



Specification	Document Number
Product Specification	DO1001 – Top Level Assembly and Specification HMX SDU1 REV A
Software	DO1003_HMX_SDU1.ino REV A
BOM	DO1002 – Bill of Materials REV B
Circuit Diagram	DO1004 – HMX SDU1 Circuit Diagram REV A
Manufacturing Procedures	SD1007 – Process Flow Diagram REV 00 SD1016 – Work Instruction, Manufacture HMX SDU1 REV 00 G-code files for 3D Printed parts define manufacturing process characteristics and are included in DO1023 REV A. Filament is defined by the part technical drawing.
Quality Inspection Procedures	100% Inspection is performed on each completed device according to the inspection-marked specifications on DO1001 REV A.
Labeling	IFU: DO1024 REV 00. This device is not a medical device. There are no regulatory or marketing labeling requirements.
Equipment/Fixture Drawing(s)	N/A

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Sterilization Requirements	☒ N/A – Non-Sterile
Expiration Date Requirements	Year(s) or ☒ N/A – no shelf life
Storage & Transport Conditions	No storage or transportation requirements.
Production Environment	No production environment requirements.

Installation, Maintenance, & Servicing Procedures	This project does not include the installation of any equipment. The equipment used in the manufacture of this device is not dedicated to this product and maintenance and servicing of it is outside the scope of this product. No service guarantees are made; post-customer support is handled under customer service. Repaired or returned devices are subject to 100% inspection before resale, no dedicated servicing procedures are required.
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Other	N/A
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Revision	Revision Date	Revision Description
00	14-AUG-2025	Initial Release

**DESIGN VERIFICATION REPORT - AMMUNITION STORAGE DISPLAY UNIT FOR
LEVER ACTION RIFLE**

DHF-[FAKEDHF#]

1. PURPOSE

This document records the design verification report for the ammunition storage display unit (Ammo SDU) HMX SDU1.

2. SCOPE

This report applies to design verification activities for the HMX SDU1. Refer to the DDP SD1001 for project and product descriptions.

3. CITED DOCUMENTS

SD1001 Design and Development Plan	REV 00
SD1002 Trace Matrix	REV 02
SD1005 Device Master Record HMX SDU1	REV 00
SD1006 Design Verification and Validation Plan	REV 00
SD1008 Design Verification Protocol HMX SDU1	REV 00

4. STUDY CONFIGURATION

The device tested is recorded in the data sheet of Attachment 1. The device tested was constructed in accordance with DMR SD1005 REV 00.

All testing was conducted according to protocol SD1008 Design Verification Protocol HMX SDU1 REV 00.

5. DEVIATIONS/EXCEPTIONAL CONDITIONS

N/A – No deviations or exceptional conditions.

6. VERIFICATION BY ANALYSIS

- 6.1** [DI-01-003] The display unit attaches to the Henry Model X using its standard top rail JDR fastener pattern on RMR Optic Mount DO1008.

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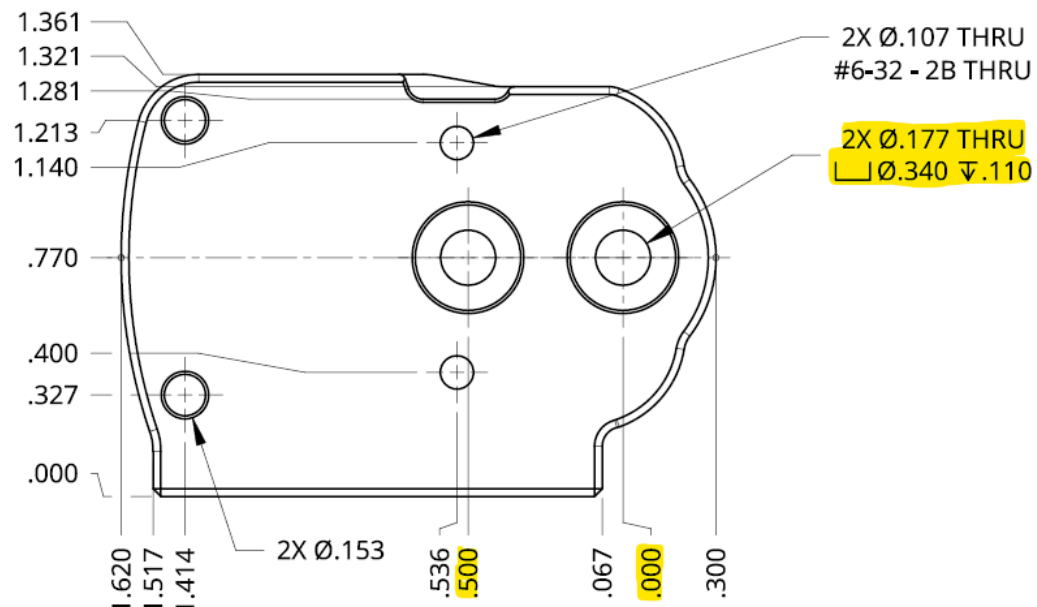


Figure 1. JDR Mount pattern on Optic mount DO1008.

6.2 [DI-03-001] The HMX SDU1 DO1001 interfaces with the magazine well and the optic mount of the Henry Model X. The magazine necessarily sits below the barrel and so does not obscure the sight line. The display unit mount DO1008 includes an RMR mount pattern for use of an RMR optic. The display unit is mounted to the left side to not obscure the optic.

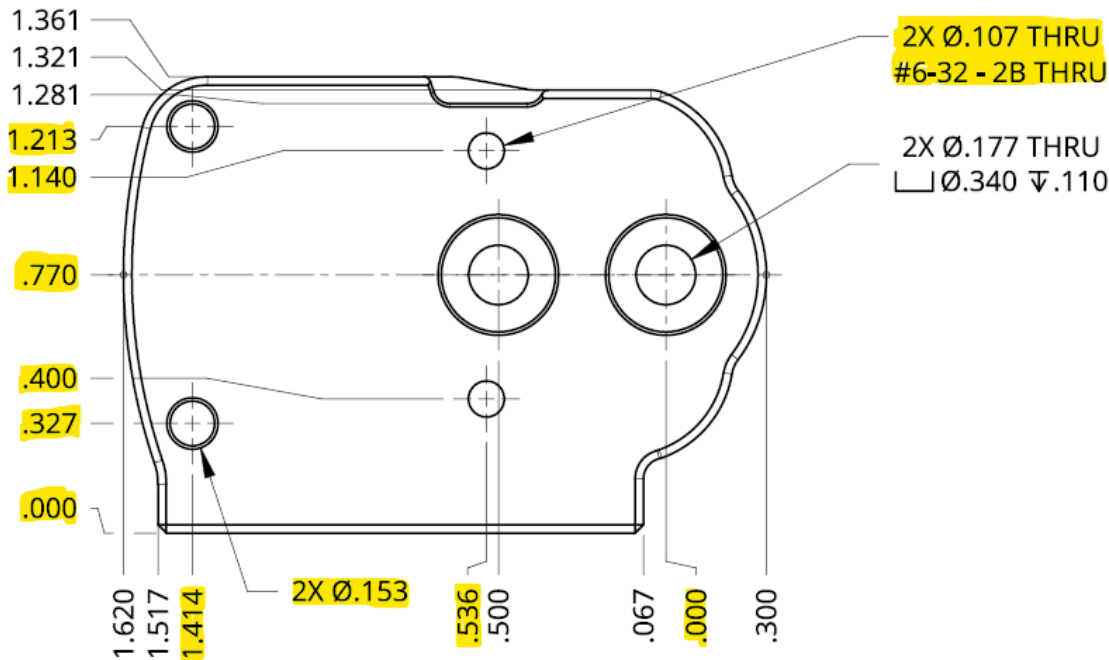


Figure 2. RMR Pattern for Optic Mount DO1008

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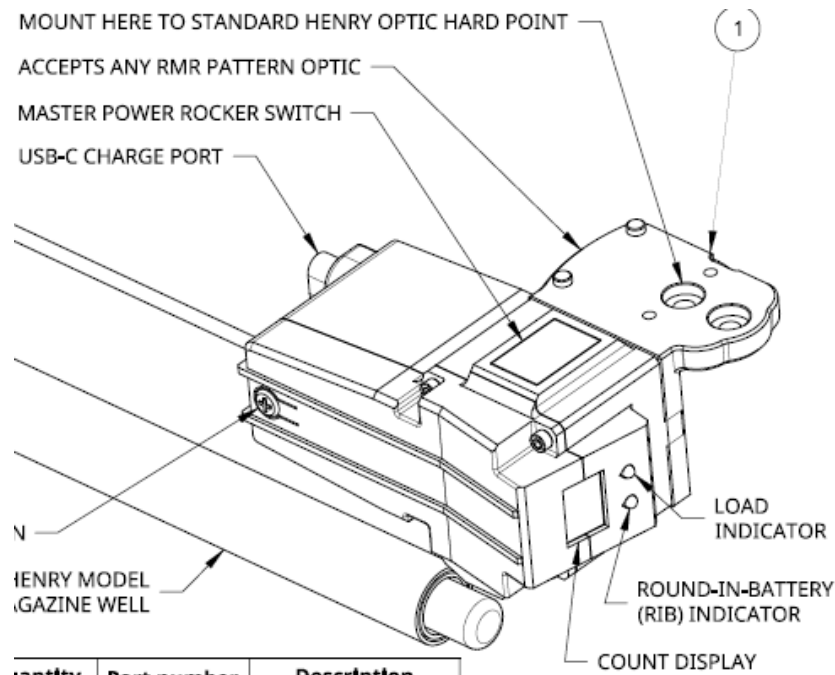


Figure 3. HMX SDU1 DO1001.

- 6.3** [DI-03-002] Per DO1007 Sensor Magazine, the sensor unit attaches to the magazine follower with the press-fit follower cable connector DO1016. This attachment is in the unused interior cavity that exists as a result of injection molding the part – it is not a functional surface. The spool cable sits coaxially inside the magazine spring, kept under tension by the lanyard spring so it remains coaxial. This way, all parts of the device stay behind the follower and out of the way of its path.

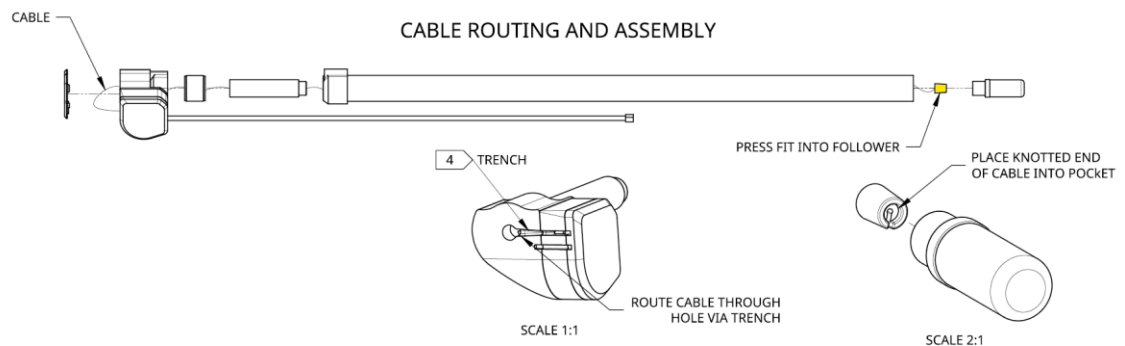


Figure 4. DO1007 Sensor magazine and follower assembly.

- 6.4** [DI-03-006] The device attaches to the Henry Model X on the top standard optic rail and by interfacing with the magazine well per top level assembly DO1001. It does not attach to or obstruct the firearm barrel.

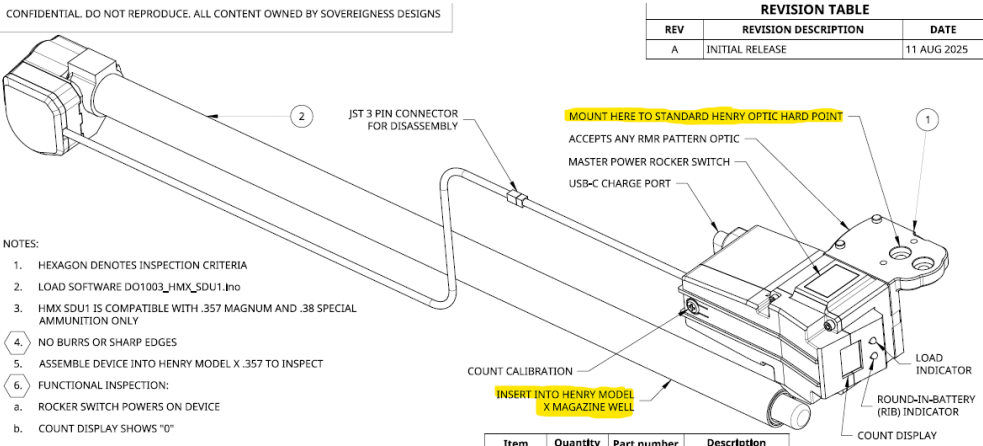


Figure 5.

6.5 [DI-03-007] The device attaches to the Henry Model X on the top standard optic rail and by interfacing with the magazine well per top level assembly DO1001. It does not attach to or interfere with the trigger or fire control group.

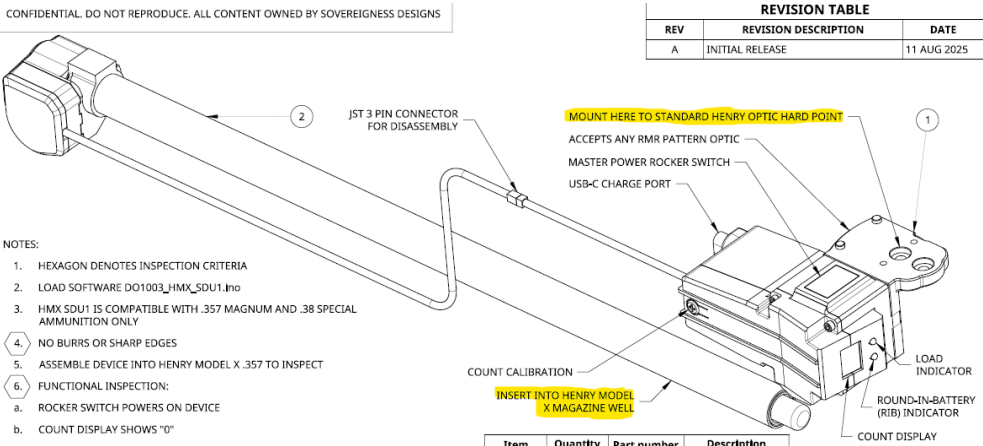
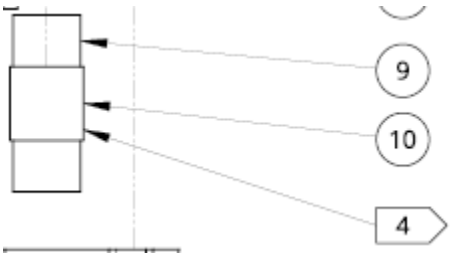


Figure 6. Interface points of the HMX SDU1 DO1001.

6.6 [DI-03-009] The device is powered by DO1002 BOM item ADAFRUIT PID 1570 3.7V LiPo 100mAh battery per DO1006. Per circuit diagram DO1004, this battery powers the Teensy 4.0 MCU input by means of the power management board BOM item ADAFRUIT PID 2124.



9	1	ADAFRUIT PID 1570	(OTS) 3.7V 100mAH LIPO BATTERY
10	1	ADAFRUIT PID 2124	(OTS) BATTERY CHARGER

Figure 7. BOM of DO1006

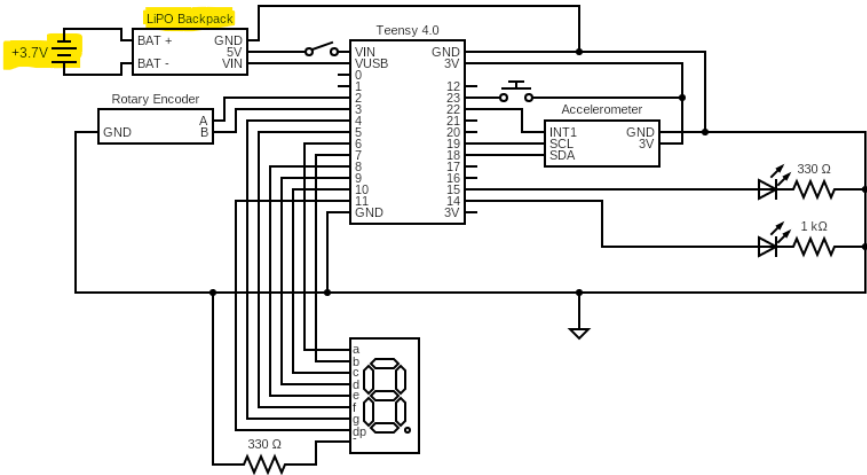


Figure 8. Circuit Diagram DO1004

6.7 [DI-04-006] Overload: The device is designed to operate with 7 rounds of ammunition. The spring spacer DO1014, among other functions, fills sufficient space such that only 7 rounds can fit inside the magazine when assembled into the Henry Model X. This is also tested by demonstration below. Jolt: Firearm recoil represents a significant source of jolt. The encoder AZ B0CF9FS3Q3 is a mechanical pulse encoder, and can send pulses up to the rate of the mechanical limit of the materials. Therefore, the limiting rate is the read speed of the MCU. Using hardware quadrature encoder functionality of the teensy 4.0 MCU, a read rate of 400Mhz can be achieved at steady state. During jolt, sudden motion of up to 1 round’s length at worst case can occur in a hundredth of a second. Each round length creates 23 pulses of the encoder. Therefore, in the worst case a jolt read rate needs to be greater than 23 pulses * 100hz * 10 discernment limit of the MCU = 23Mhz. Software DO1003 runs the device at 400Mhz, so the encoder will operate even in jolt conditions. Count loss: As a firearm, the user has a large amount of direct control with the loaded state of the magazine. Therefore, the most reliable method to handle count loss is to give the user controls for recovering in the event of lost count. The reset button of DO1006 will calibrate the device to zero rounds if the magazine is empty but the display shows a number greater than zero. A slow double press will calibrate the count to 7 if the device is turned on in the fully loaded state. See software DO1003. The user can therefore put the firearm in its two most common loaded states (empty or full) can calibrate the device to match. Connection loss: The circuit diagram of DO1004 and software of DO1003 do not require the encoder to be connected at all times to function – if the encoder connector is disconnected and reconnected during use, from the point of view of the MCU nothing has occurred. If the magazine were to change

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state while the encoder connector was disconnected, this would result in count loss which must be handled with re-calibration, but when reconnected and recalibrated the device returns to normal operating function.

- 6.8** [DI-05-001] The device display is vertically oriented and position to the left side of the firearm, below the optic. This means it is upright in a readable position 2 inches to the left and half an inch below the sight line in a shooting position – this is well within the viewable angle.

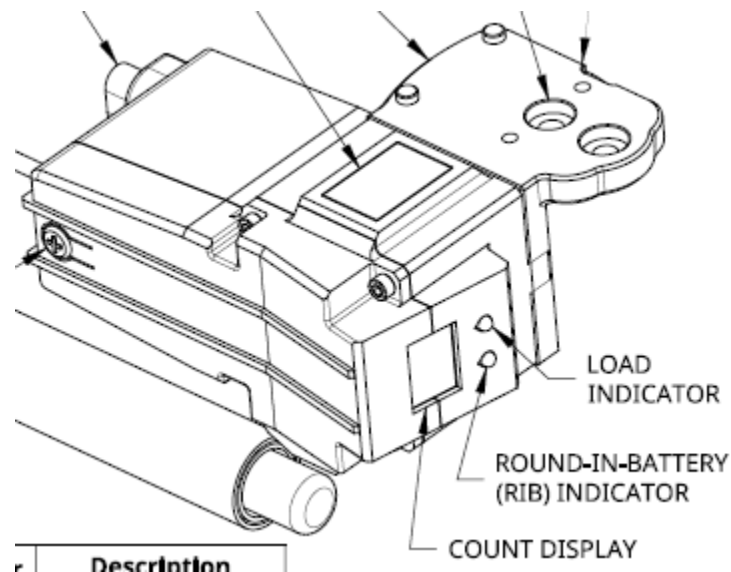
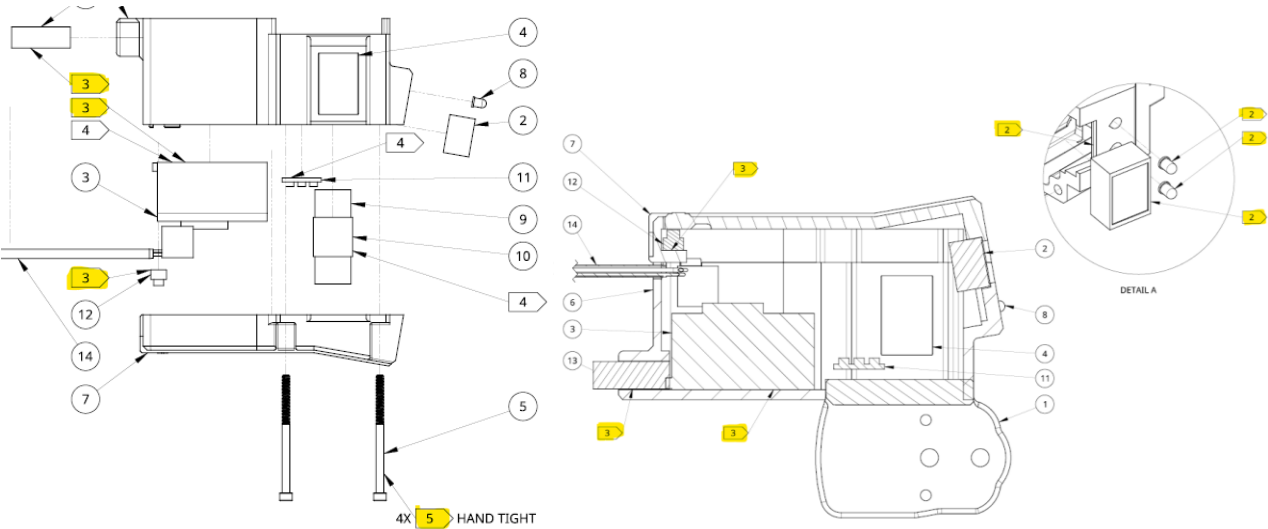


Figure 9. DO1001 Display Unit.

- 6.9** [DI-06-001, DI-06-002] The device contains two enclosures of components not intended to be externally exposed: The electrical components inside the Housing DO1009 in assembly DO1006; and the mechanical components inside the sensor unit (Not the magazine) DO1007. The other components do not have spaces for rain or dust to ingress. The assembly of the Housing in assembly DO1006 uses cyanoacrylate and hot melt glue to fill gaps and permanently enclose the electronics. There is only one gap in the housing, at the reset button, but it is sealed on the interior with hot melt glue. In the sensor unit DO1007, the spool case cover and sensor unit cover enclose the spool case and tube mount and are sealed with glue. There are no plausible pathways for rainwater or dust ingress.



2.

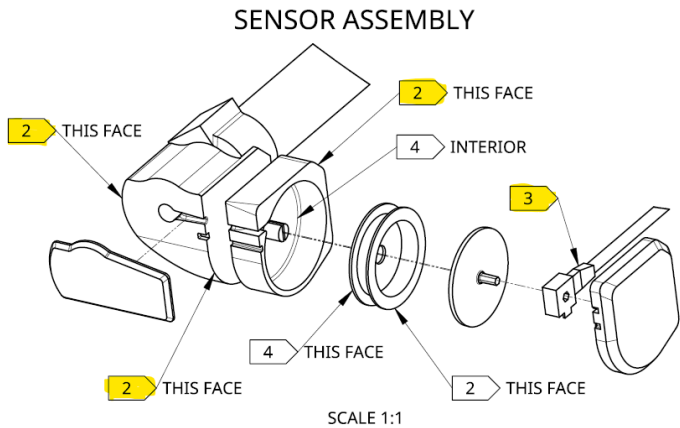
APPLY OTS LOCTITE 1647358 CYANOCRYLATE GLUE TO ASSEMBLE
3.

APPLY OTS HOT-MELT GLUE TO ASSEMBLE
4.

COVER CONTACTS WITH OTS ADAFRUIT PID 3057 NYLON TAPE AFTER SOLDERING
5.

APPLY OTS LOCTITE BLUE 242 THREADLOCKER

Figure 10. DO1006 Sealing.



2.

APPLY OTS LOCTITE 1647358 CYANOCRYLATE GLUE TO ASSEMBLE
3.

APPLY OTS HOT-MELT GLUE TO ASSEMBLE

Figure 11. DO1007 Sealing.

Page 1 of 12	Document #:	SD1009
	Revision #:	00

7. TEST RESULTS

The raw data is captured in Attachment 1 Complete Data Sheet. Table 1 summarizes the results of the tests and comparison to acceptance criteria and acceptance or rejection of each result.

Acceptance criteria are released under SD1006 DV&V Plan.

Table 1.

ID#	Requirement	Test Protocol Step	Acceptance Criteria	Expected Result	Actual Result	Pass / Fail	Comments
DI-01-001	The device shall interface with the magazine well of the Henry Model X .357 in stock condition.	8.1	The device magazine fits and locks into the Henry Model X .357 magazine well.	YES	YES	PASS	N/A
DI-01-002	The device shall operate with .357 and .38 special ammunition.	8.11 9.11	The device meets all test and demonstration requirements with both .357 and .38 special ammunition.	YES	YES	PASS	N/A
DI-02-001	The magazine end of the device shall connect and disconnect with the firearm identically to the stock magazine.	8.1	The device magazine shall secure with the same method as the stock magazine.	YES	YES	PASS	N/A
DI-02-002	The device shall mount to standard m-lok, picatinny, or JDR pattern mounts using removable fasteners.	8.3 8.13	The device can be removed after installation.	YES	YES	PASS	N/A
DI-03-003	With the device installed, 7 rounds shall be able to be stored in the magazine tube.	8.7	Seven rounds stored in the magazine tube.	7	7	PASS	N/A

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	Revision #:	00

DI-03-004	With the device installed, a cyclic fire rate of at most 3 seconds per round shall be achievable.	9.7	A cyclic fire rate less than or equal to 3 seconds per round is achieved in each of 7 trials (one full magazine).	≤ 3 seconds / round	2.9 seconds / round	PASS	N/A
DI-03-005	The device shall not obstruct the bolt and ejection window.	8.10	The firearm is able to fire and eject spent casings with the device installed.	YES	YES	PASS	N/A
DI-03-008	The device shall not obstruct the charging lever.	8.9	The charging lever can be fully extended and retracted with the device installed.	YES	YES	PASS	N/A
DI-03-010	The device shall not obstruct or interfere with the trigger safety or lever-deactivator.	8.9	The lever deactivator is functional with the device installed.	YES	YES	PASS	N/A
DI-04-001	The device shall be capable of displaying the numerals 0-7 visibly in daylight conditions.	8.7	The device display is visible in outdoor daylight conditions.	YES	YES	PASS	N/A
DI-04-002	The device shall display 0 when the magazine is empty.	8.10	After calibration, the device displays zero when the magazine is unloaded.	YES	YES	PASS	N/A
DI-04-003	The device shall increment each time a round is loaded into the magazine.	8.7	Loading the firearm fully from the unloaded condition increments the display from 0 to 7.	YES	YES	PASS	N/A
DI-04-004	The device shall decrement each time a round is removed from the magazine.	8.10	Clearing the firearm from the fully loaded	YES	YES	PASS	N/A

Page 3 of 12	Document #:	SD1009
	Revision #:	00

			condition decrements the display from 7 to 0.				
DI-04-005	The device shall have an indicator for round-in-battery.	9.6 9.7	The round-in-battery indicator red LED lights when a round is loaded and turns off when fired.	YES	YES	PASS	N/A
DI-04-006	The device shall have error handling for over-load, jolt, connection loss and count loss.	8.8	Eighth round does not fit into magazine and count remains at 7.	YES	YES	PASS	N/A
DI-04-007	The device shall have manual controls for resetting and maxing the count.	8.6	Pressing the reset button changes the count to zero when the count is not zero and sets the count to seven when it is zero.	YES	YES	PASS	N/A
DI-04-008	The device shall maintain the appropriate count when the firearm is fired.	9.7	Firing the firearm does not change the displayed count in 7 trials (one full magazine).	YES	YES	PASS	N/A
DI-04-009	The device shall be operable continuously for at least 1 hour.	8.12	Time to battery shut-off when on exceeds 1 hour.	≥ 1 hour	2.2 hours	PASS	N/A
DI-05-002	The display shall be viewable with the firearm shouldered and standard sight picture acquired.	9.4	The display can be viewed	YES	YES	PASS	N/A
DI-05-003	The display shall be mounted on the opposite side of the firearm as the ejection port.	7.1	The device display is on the left side of the firearm.	YES	YES	PASS	N/A

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	Revision #:	00

DI-05-004	The device shall increase the weight of the firearm by less than or equal to 3 lbs.	7.2	The device components weigh less than or equal to 3 lbs.	≤ 3 lbs.	1.5 lbs.	PASS	N/A
DI-05-005	The device controls shall be operable with a single hand.	9.5	The power switch and reset button can be actuated with one hand while holding the firearm.	YES	YES	PASS	N/A
DI-05-006	The device shall not obstruct the handle or fore-grip.	9.4	The handle and foregrip can be accessed with the device installed.	YES	YES	PASS	N/A

8. CONCLUSION

Per Table 1, all acceptance criteria are met. No exceptional conditions or deviations are recorded.

The product HMX SDU1 built to DMR SD1005 REV 00 satisfies all design requirements listed in Trace Matrix SD1002 REV 02.

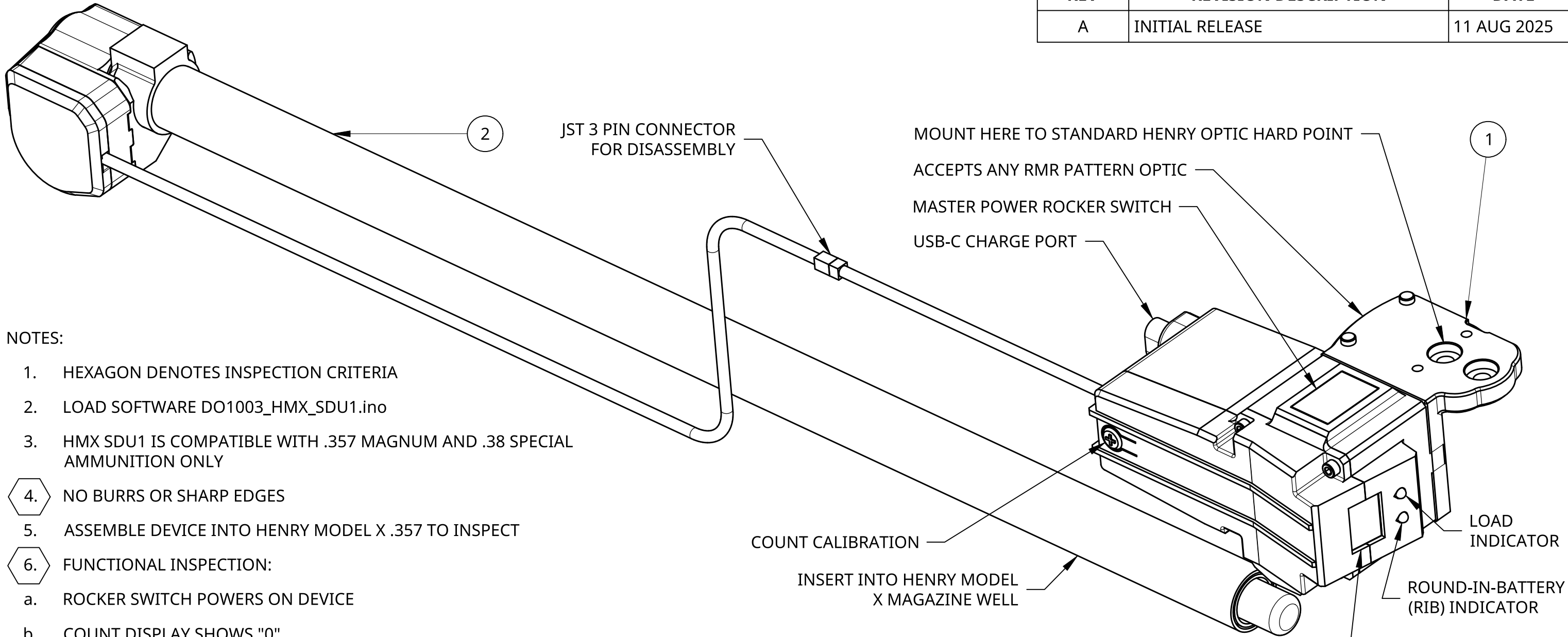
9. ATTACHMENTS

Attachment 1: Complete Data Sheet

Page 1 of 12	Document #:	SD1009
	Revision #:	00

CONFIDENTIAL. DO NOT REPRODUCE. ALL CONTENT OWNED BY SOVEREIGNESS DESIGNS

REVISION TABLE		
REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	11 AUG 2025



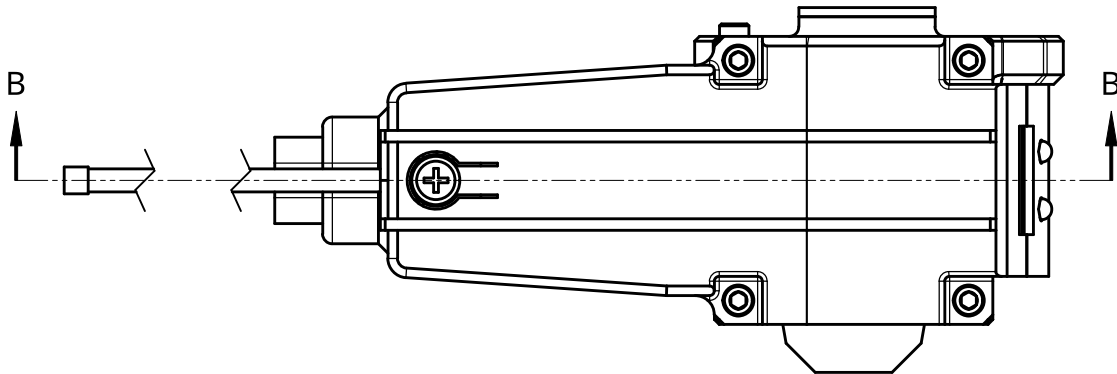
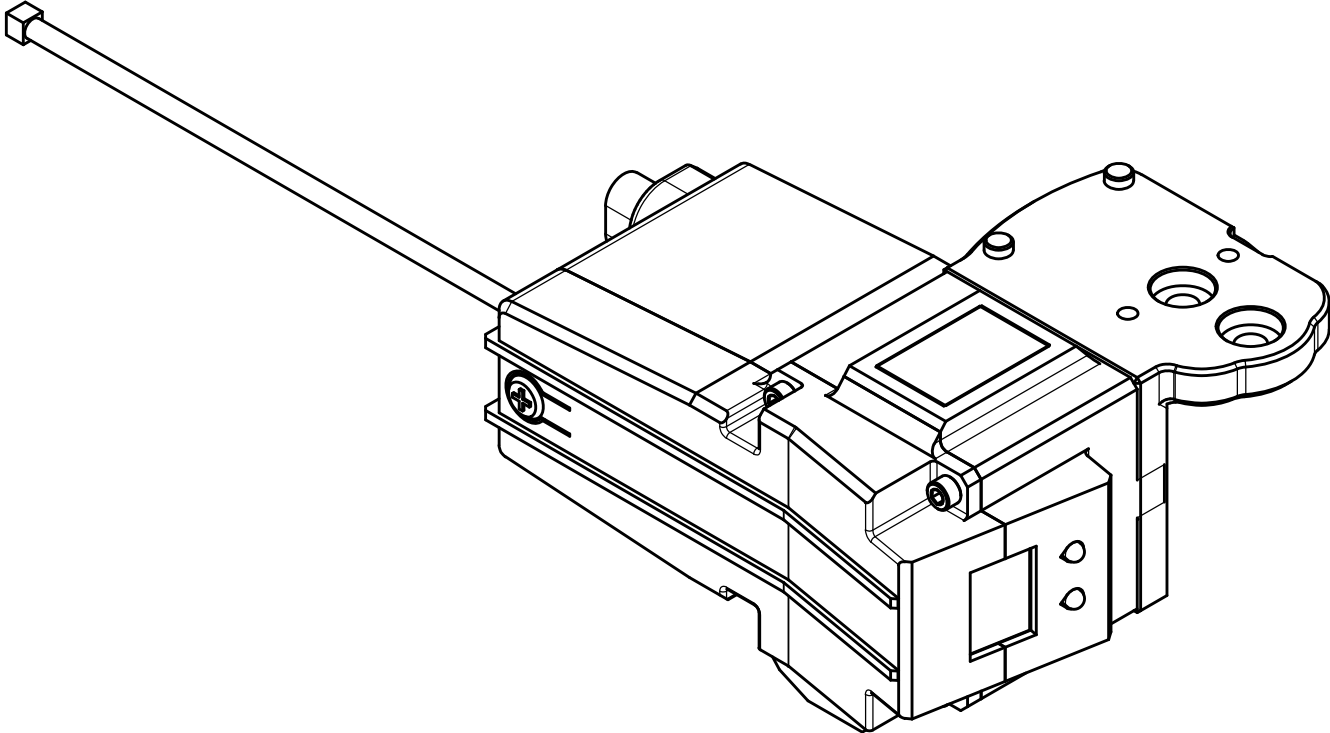
Item	Quantity	Part number	Description
1	1	DO1006	DISPLAY UNIT
2	1	D01007	SENSOR MAGAZINE

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32	NAME		DATE	TITLE HMX SDU1				
	DRAWN		JESSICA BOYERS					8 AUG 2025
DO NOT SCALE DRAWING					HMX SDU1			
THIRD ANGLE PROJECTION	MATERIAL		FINISH		SIZE	DWG NO.	REV.	
	N/A		N/A		B	DO1001	A	
					SCALE	1:1	SHEET	
							1 of 1	

CONFIDENTIAL. DO NOT REPRODUCE. ALL CONTENT OWNED BY SOVEREIGNESS DESIGNS

REVISION TABLE

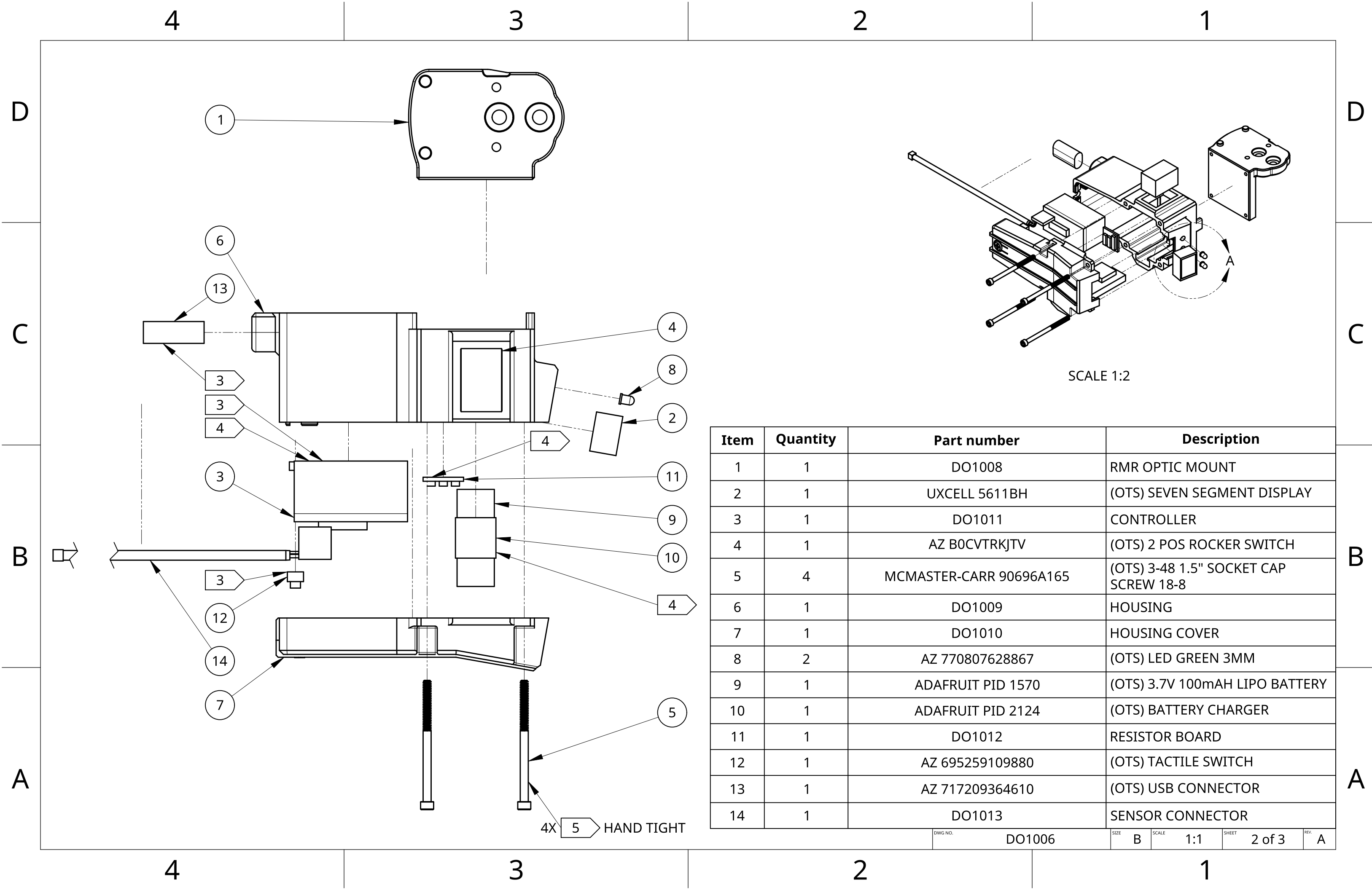
REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	26 MAY 2025



NOTES:

1. WIRING NOT SHOWN. SOLDER CONNECTIONS ACCORDING TO CIRCUIT DIAGRAM DO1004 USING OTS $\frac{60}{40}$ SN-PB SOLDER.
2. APPLY OTS LOCTITE 1647358 CYANOCRYLATE GLUE TO ASSEMBLE
3. APPLY OTS HOT-MELT GLUE TO ASSEMBLE
4. COVER CONTACTS WITH OTS ADAFRUIT PID 3057 NYLON TAPE AFTER SOLDERING
5. APPLY OTS LOCTITE BLUE 242 THREADLOCKER
6. OTS DESIGNATES OFF-THE-SHELF COMPONENTS THAT DO NOT HAVE DETAILED DRAWINGS AND MAY BE REPRESENTED WITH SIMPLIFIED GEOMETRY WHERE APPLICABLE. REFER TO THE SUPPLIER PRODUCT SPECIFICATIONS.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32		NAME	DATE	TITLE DISPLAY UNIT		
	DRAWN	JESSICA BOYERS	14 MAY 25			
DO NOT SCALE DRAWING				SIZE	DWG NO.	REV.
THIRD ANGLE PROJECTION	MATERIAL	FINISH		B	DO1006	A
	N/A	N/A		SCALE	1:1	SHEET
						1 of 3



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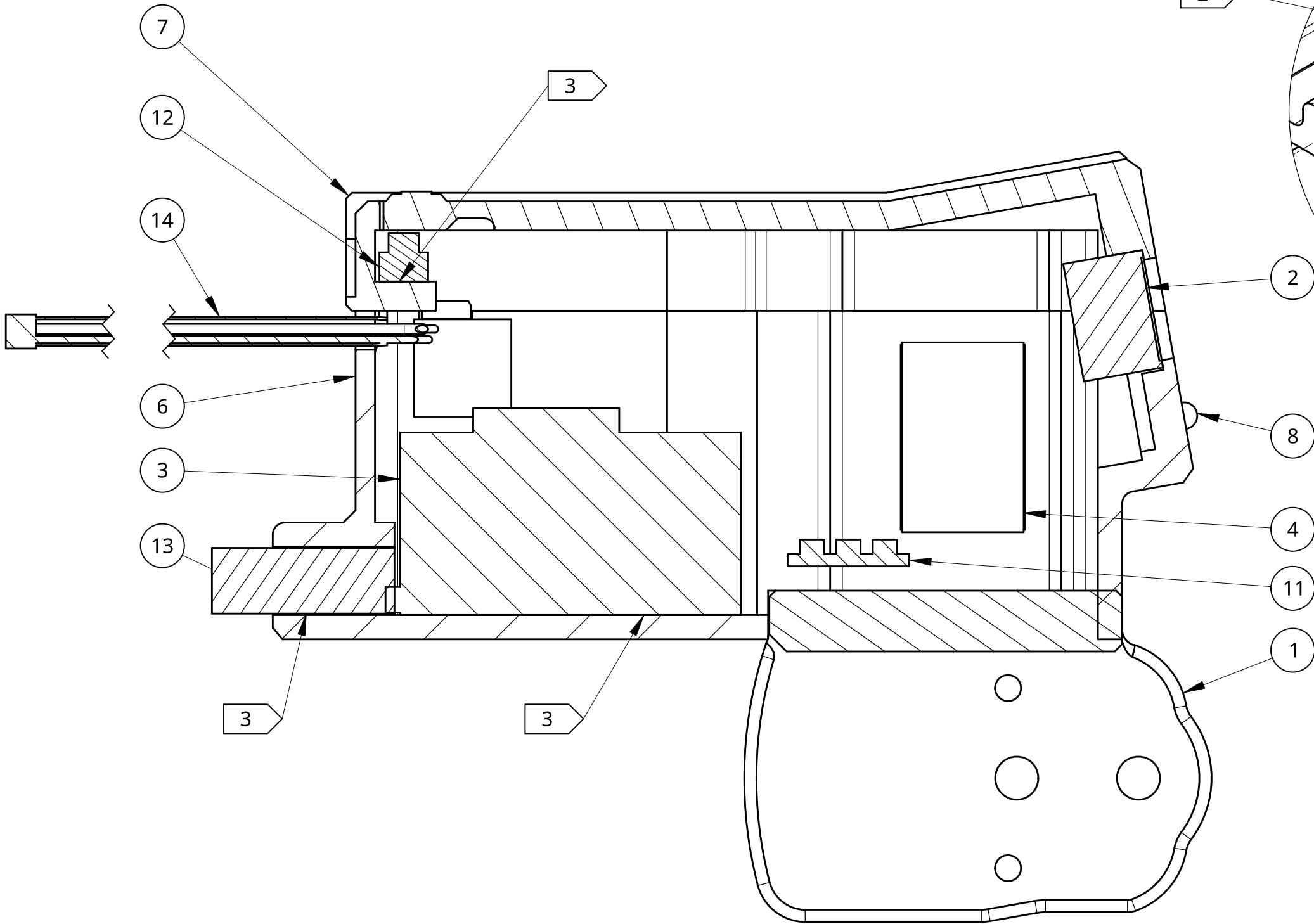
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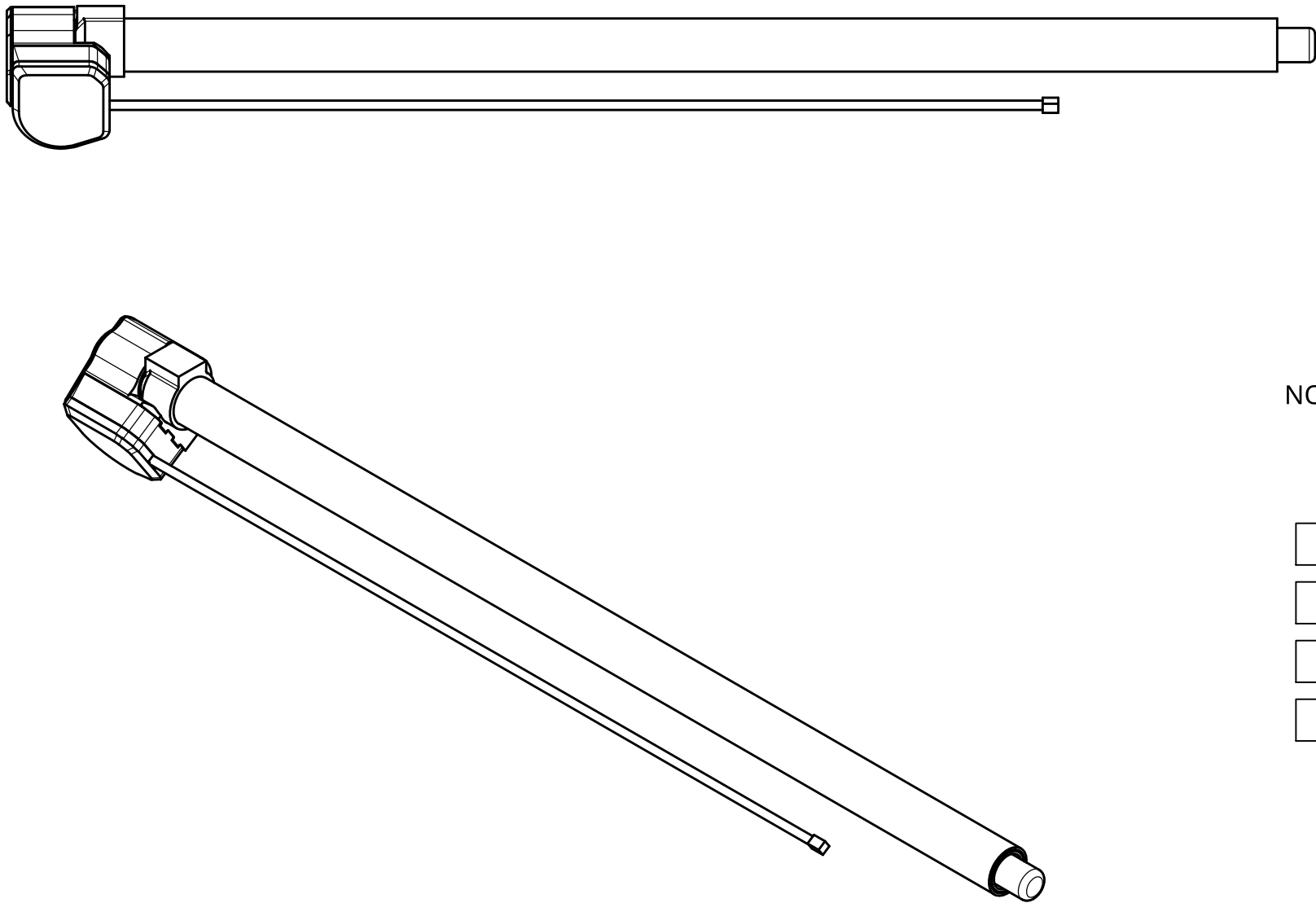
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REVISION TABLE		
REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	10 AUG 2025
B	ADDED SPRING SPACER	12 NOV 2025



NOTES:

1.

WIRING NOT SHOWN. SOLDER CONNECTIONS ACCORDING TO CIRCUIT DIAGRAM DO1004 USING OTS $\frac{60}{40}$ SN-PB SOLDER.

2.

APPLY OTS LOCTITE 1647358 CYANOCRYLATE GLUE TO ASSEMBLE

3.

APPLY OTS HOT-MELT GLUE TO ASSEMBLE

4.

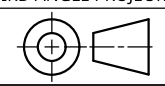
APPLY OTS WD-40 SPECIALIST DRY LUBE 1 COAT

5.

APPLY OTS LOCTITE BLUE 242 THREADLOCKER

6.

OTS DESIGNATES OFF-THE-SHELF COMPONENTS THAT DO NOT HAVE DETAILED DRAWINGS AND MAY BE REPRESENTED WITH SIMPLIFIED GEOMETRY WHERE APPLICABLE. REFER TO THE SUPPLIER PRODUCT SPECIFICATIONS.

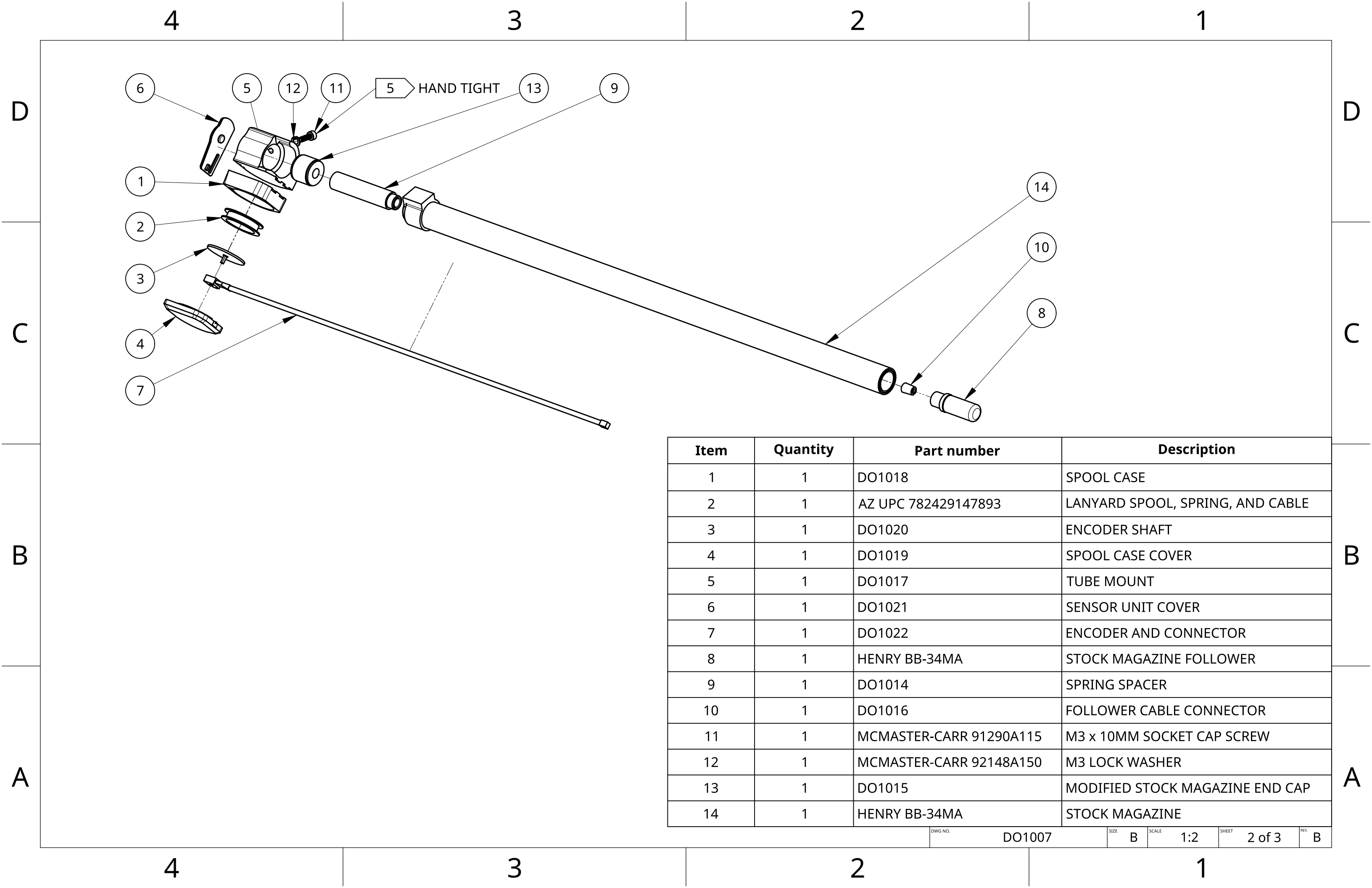
UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32		NAME	DATE	TITLE SENSOR MAGAZINE		
	DRAWN	JESSICA BOYERS	5 JUL 2025			
	DO NOT SCALE DRAWING				SIZE	DWG NO.
THIRD ANGLE PROJECTION	MATERIAL	FINISH		B	DO1007	B
	N/A	N/A		SCALE	1:2	SHEET
						1 of 3

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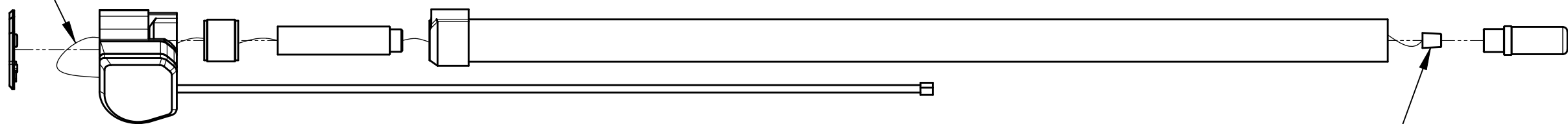
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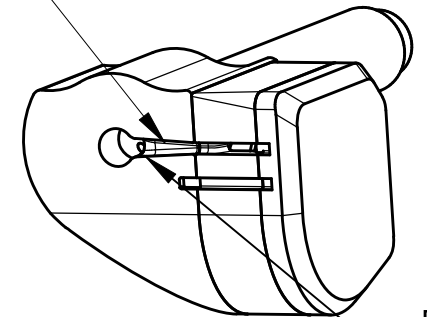
CABLE ROUTING AND ASSEMBLY

CABLE



4

TRENCH

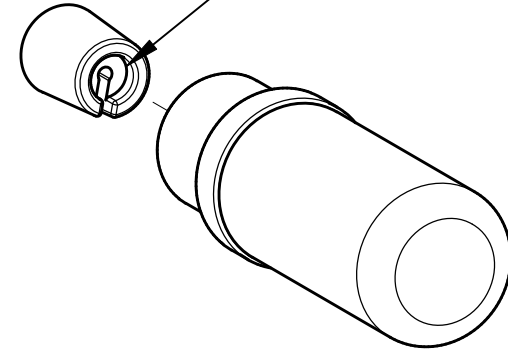


SCALE 1:1

ROUTE CABLE THROUGH
HOLE VIA TRENCH

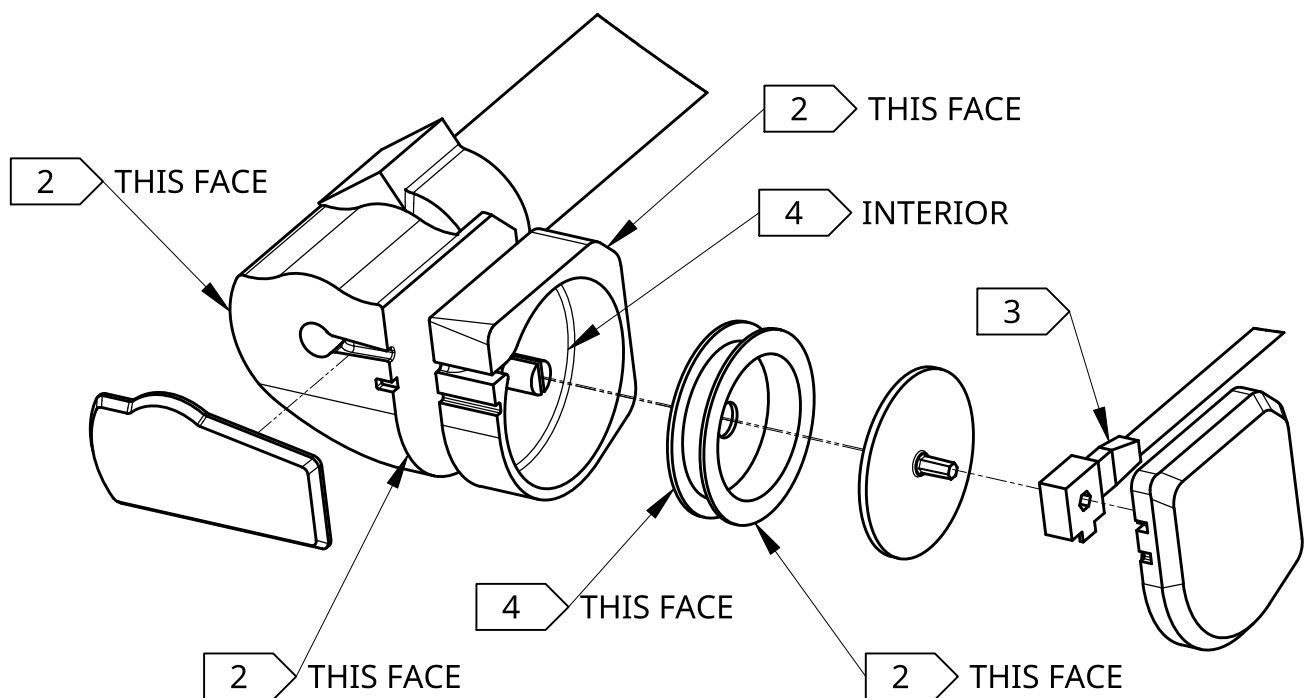
PRESS FIT INTO FOLLOWER

PLACE KNOTTED END
OF CABLE INTO POCKET



SCALE 2:1

SENSOR ASSEMBLY



SCALE 1:1

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REVISION TABLE

REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	2 JUN 2025

2X Ø.079 ∇ .250
#3-48 - 2B ∇ .250

2X Ø.079 THRU
#3-48 - 2B THRU

.050 X 45.0°

1.361

1.321

1.281

1.213

1.140

.770

.400

.327

.000

1.620
1.517
1.414

2X Ø.153

.536
.500

.067
.000

.300

2X Ø.107 THRU
#6-32 - 2B THRU

2X Ø.177 THRU
 $\sqcap$ Ø.340 ∇ .110

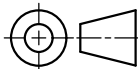
UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES

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.XX = $\pm$.01
.XXX = $\pm$.001
.XXXX = $\pm$.0005

ANGULAR = $\pm$ 1°
FRACTIONAL = $\pm$ 1/32

DO NOT SCALE DRAWING

THIRD ANGLE PROJECTION

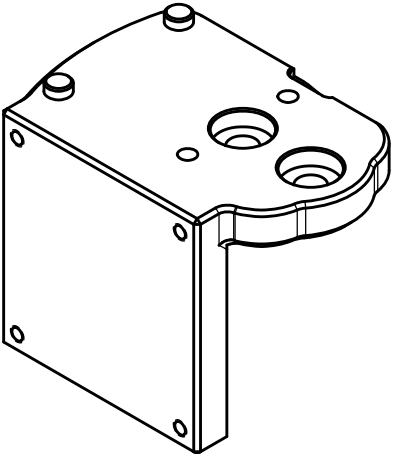


	NAME	DATE
DRAWN	JESSICA BOYERS	14 MAY 25

TITLE		
RMR OPTIC MOUNT		

MATERIAL	FINISH
6061 Aluminum	Anodized (Blue)

SIZE	DWG NO.	REV.
B	DO1008	A
SCALE	1 of 2	
2:1		



SCALE 1:1

- NOTES:
- BREAK ALL SHARP EDGES, REMOVE ALL BURRS
 - SURFACE ROUGHNESS - TOOL MARKS REMOVED

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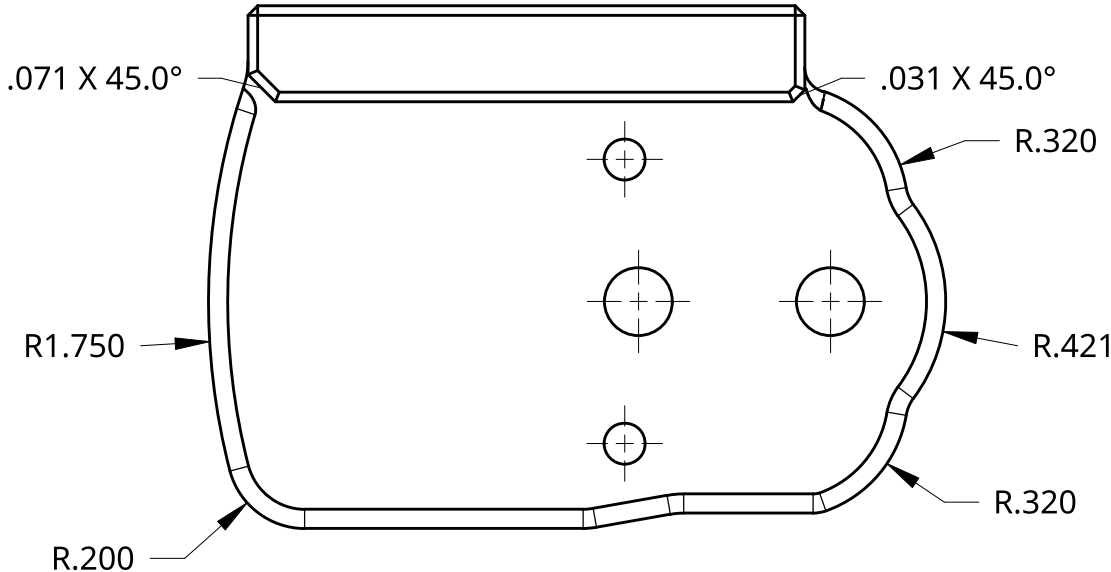
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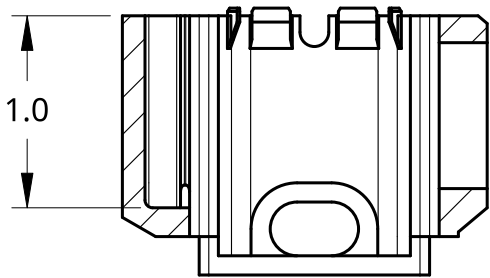
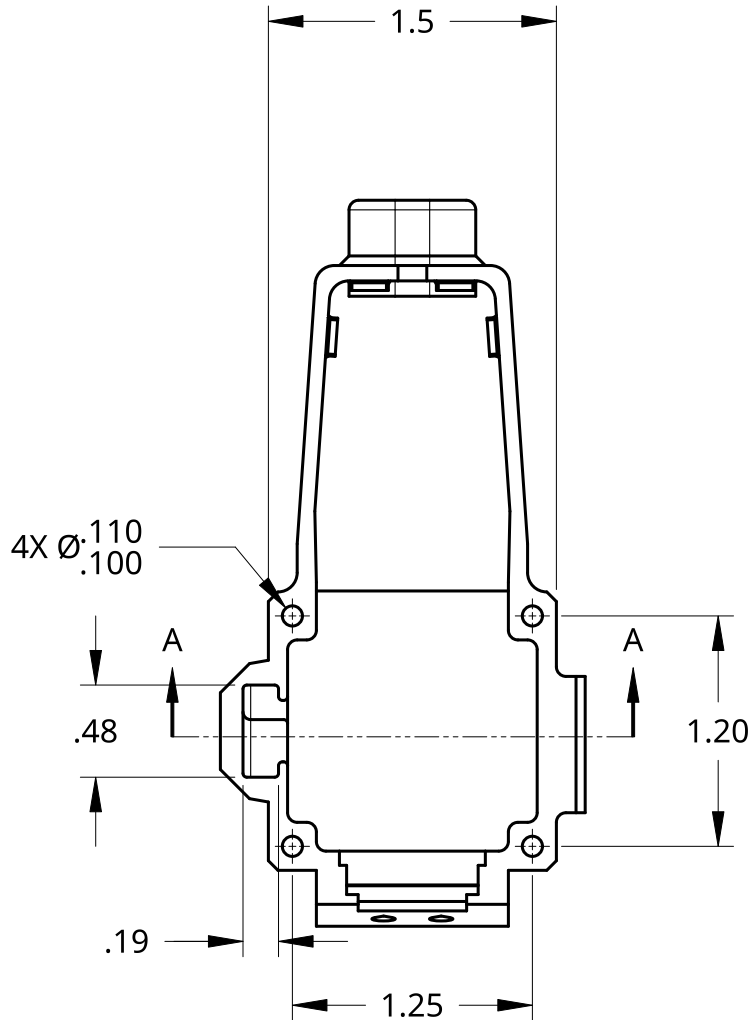
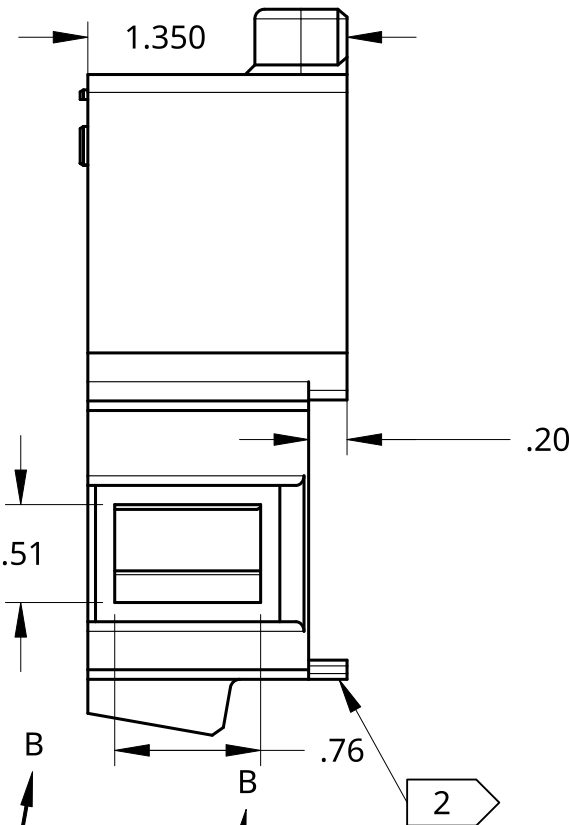
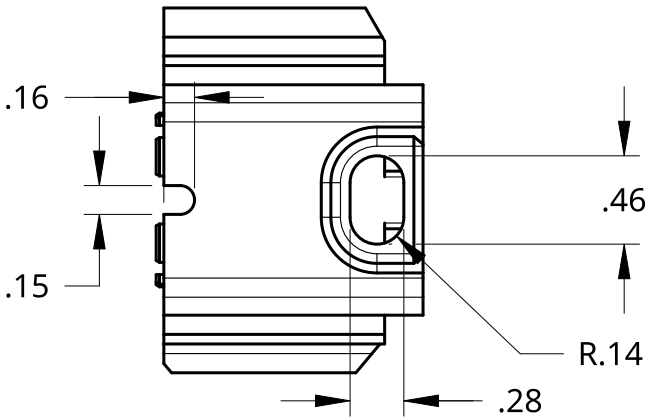
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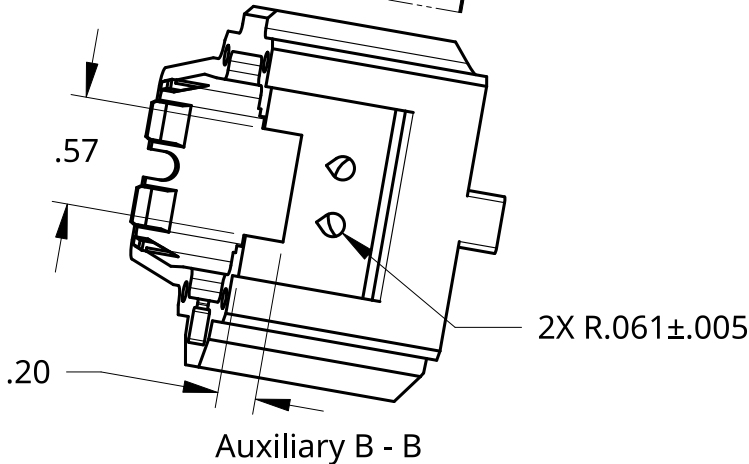
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REVISION TABLE

REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	28 MAY 2025



SECTION A - A



Auxiliary B - B

- NOTES:
- 1. THIS PART IS 3D PRINTED. THE SOLID MODEL COMPRISES A COMPLETE DESIGN DOCUMENT. THIS DOCUMENT HAS ADDITIONAL SPECIFICATIONS AND INSPECTION CRITERIA.
 - 2. THIS TAB FOR 3D PRINTING SUPPORT, REMOVE AFTER PRINT.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32 DO NOT SCALE DRAWING THIRD ANGLE PROJECTION		NAME	DATE	TITLE HOUSING		
	DRAWN	JESSICA BOYERS	14 MAY 25			
	MATERIAL	PLA	FINISH	AS PRINTED	SIZE B	DWG NO. DO1009
					SCALE 1:1	REV. A
					SHEET 1 of 1	

4

3

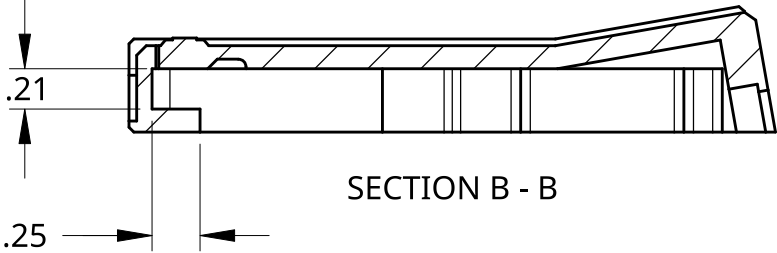
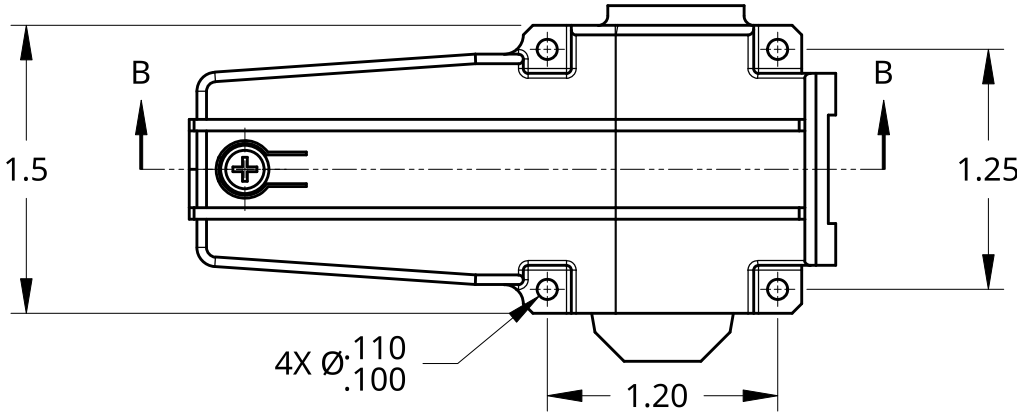
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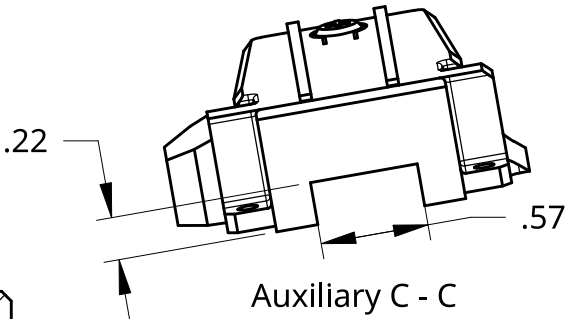
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REVISION TABLE

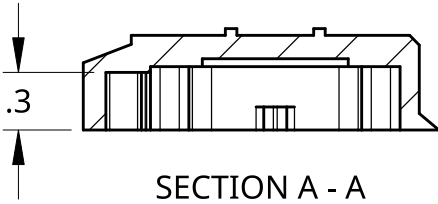
REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	28 MAY 2025



SECTION B - B

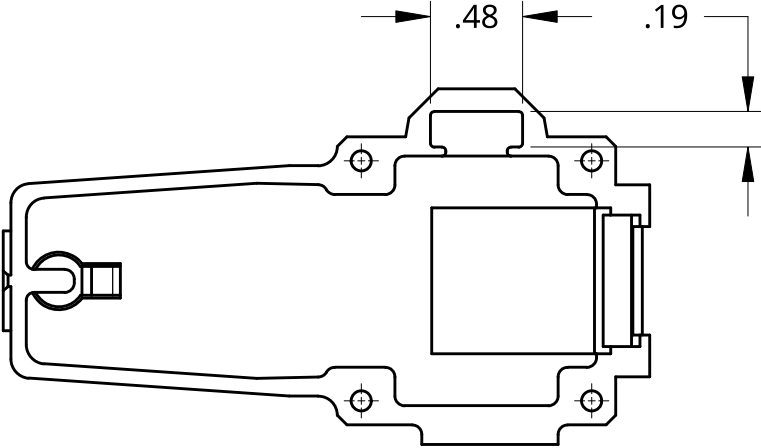


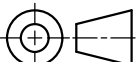
Auxiliary C - C



SECTION A - A

- NOTES:
- THIS PART IS 3D PRINTED. THE SOLID MODEL COMPRISES A COMPLETE DESIGN DOCUMENT. THIS DOCUMENT HAS ADDITIONAL SPECIFICATIONS AND INSPECTION CRITERIA.



UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32	NAME		DATE	TITLE HOUSING COVER				
	DRAWN		JESSICA BOYERS				14 MAY 25	
DO NOT SCALE DRAWING								
THIRD ANGLE PROJECTION	MATERIAL		FINISH		SIZE	DWG NO.	REV.	
	PLA		AS PRINTED		B	DO1010	A	
				SCALE	1:1		SHEET	1 of 1

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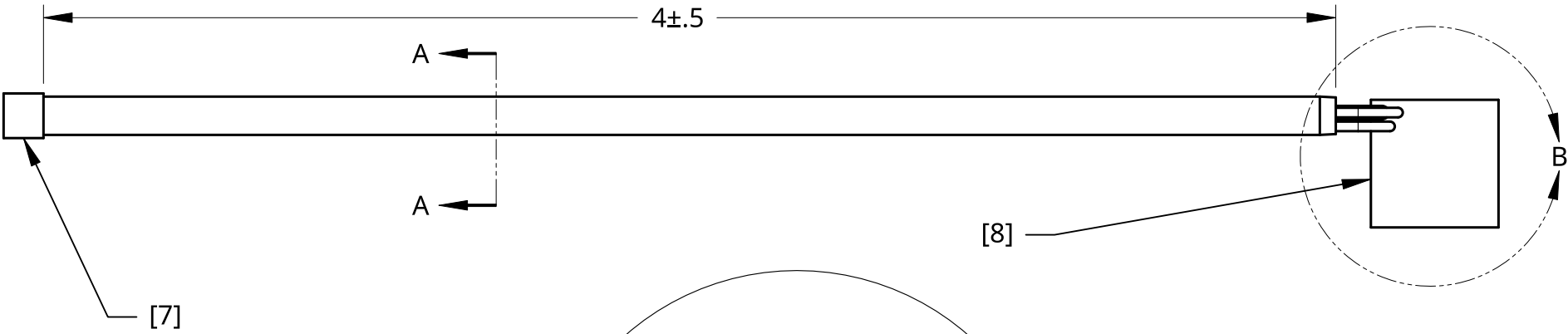
D

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SOLDER TO STANDARD HEADER PIN (FEMALE)

DETAIL B
SCALE 4:1



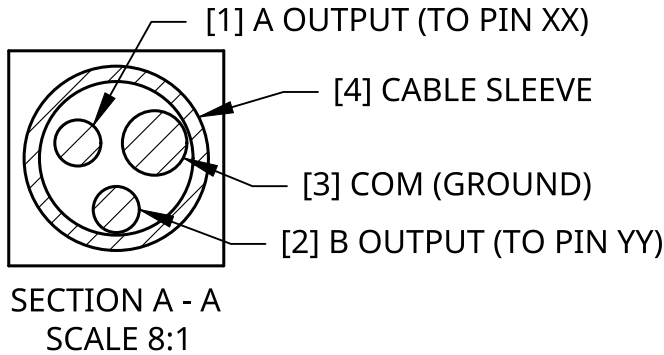
NOTES:

1. [6] HEATSHRINK TUBING OVER NYLON TAPE THIS AREA

2. OTS DESIGNATES OFF-THE-SHELF COMPONENTS THAT DO NOT HAVE DETAILED DRAWINGS AND MAY BE REPRESENTED WITH SIMPLIFIED GEOMETRY WHERE APPLICABLE. REFER TO THE SUPPLIER PRODUCT SPECIFICATIONS.

REVISION TABLE

REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	26 MAY 2025



BOM

ITEM	QUANTITY	PART NUMBER	PART DESCRIPTION
1	1	N/A	(OTS) 30 GAUGE INSULATED WIRE WRAP WIRE
2	1	N/A	(OTS) 30 GAUGE INSULATED WIRE WRAP WIRE
3	1	N/A	(OTS) 24 GAUGE SOLID COPPER WIRE
4	1	AZ B08333J864	(OTS) BRAIDED CABLE SLEEVE, 4"
5	2	ADAFRUIT PID 3057	(OTS) NYLON TAPE
6	2	N/A	(OTS) HEAT SHRINK TUBING .1" RECOVERED
7	1	AZ 602938070973	(OTS) JST SM 3 MALE CONNECTOR
8	1	SPARKFUN PRT-22892	(OTS) FEMALE HEADER PIN CONNECTOR 5 PIN

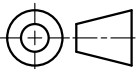
UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES

.X = ±.1
.XX = ±.01
.XXX = ±.001
.XXXX = ±.0005

ANGULAR = ±1°
FRACTIONAL = ±1/32

DO NOT SCALE DRAWING

THIRD ANGLE PROJECTION



NAME

DATE

DRAWN

JESSICA BOYERS

14 MAY 25

TITLE

SENSOR CONNECTOR, DISPLAY SIDE

SIZE
B

DWG NO.

DO1013

REV.

A

SCALE
2:1

SHEET

1 of 1

4

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2

1

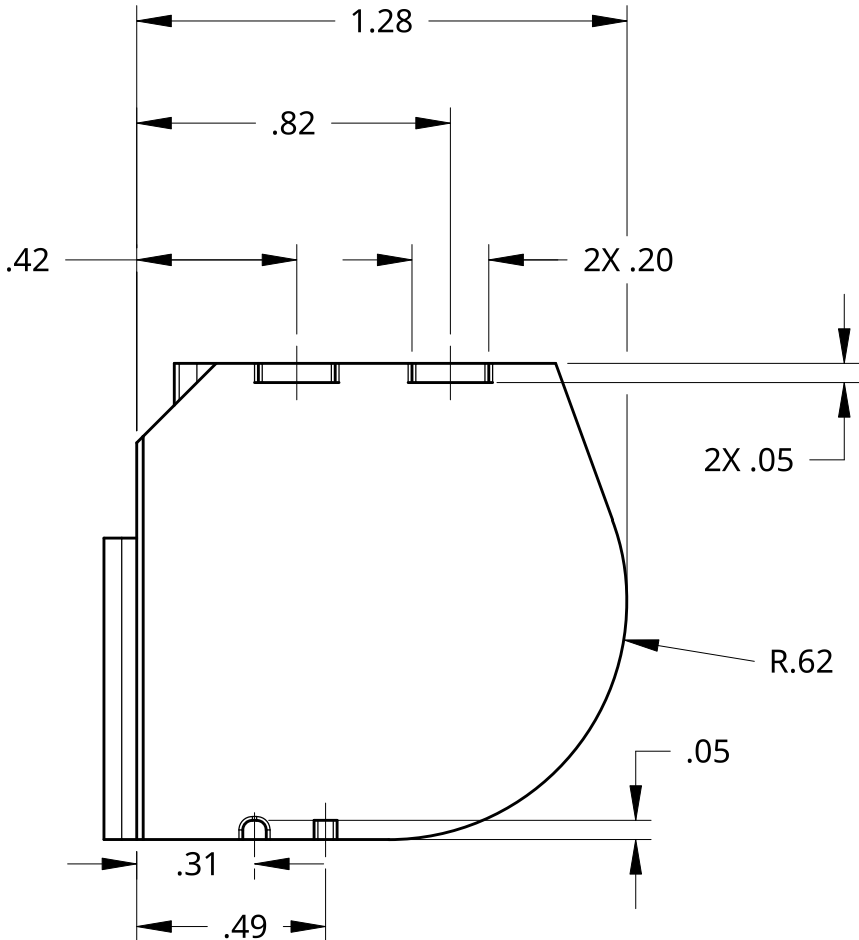
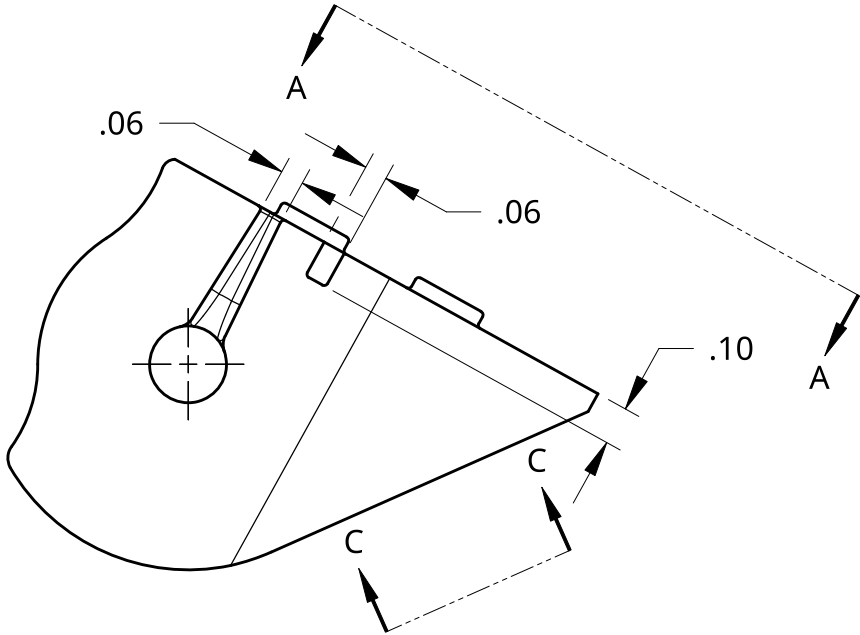
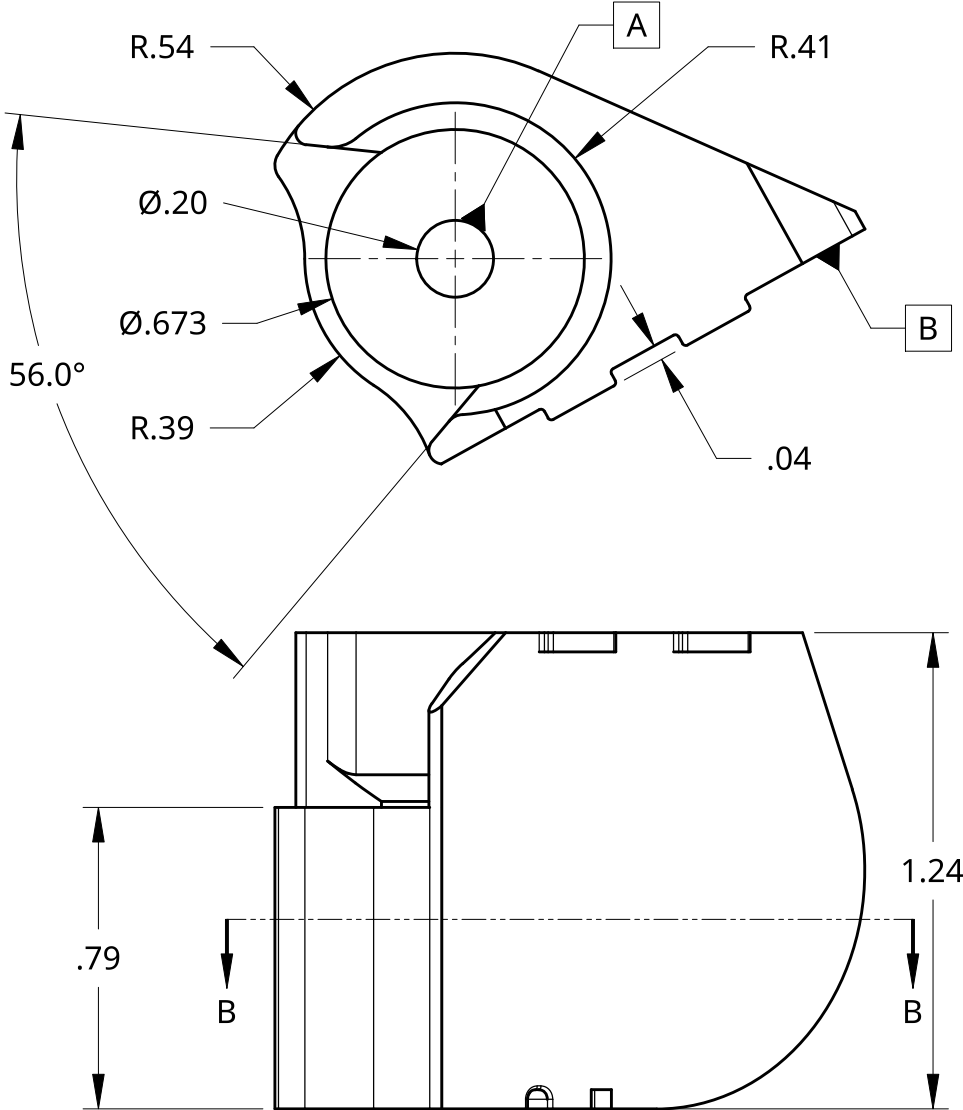
D

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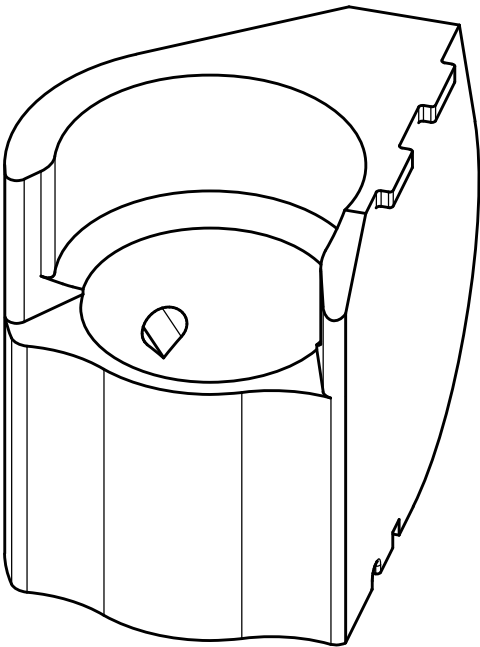
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Auxiliary A - A

NOTES:

1. THIS PART IS 3D PRINTED. THE SOLID MODEL COMPRISES A COMPLETE DESIGN DOCUMENT. THIS DOCUMENT HAS ADDITIONAL SPECIFICATIONS AND INSPECTION CRITERIA.



D

C

B

A

REVISION TABLE

REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	16 JUL 2025

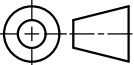
UNLESS OTHERWISE SPECIFIED,
DIMENSIONS ARE IN INCHES

.X = ±.1
.XX = ±.01
.XXX = ±.001
.XXXX = ±.0005

ANGULAR = ± 1°
FRACTIONAL = ±1/32

DO NOT SCALE DRAWING

THIRD ANGLE PROJECTION



NAME

DATE

DRAWN

JESSICA BOYERS

15 JUL 2025

TITLE

TUBE MOUNT

SIZE

DWG NO.

B

DO1017

REV.

A

SCALE

2:1

SHEET

1 of 2

4

3

2

1

4

3

2

1

D

D

C

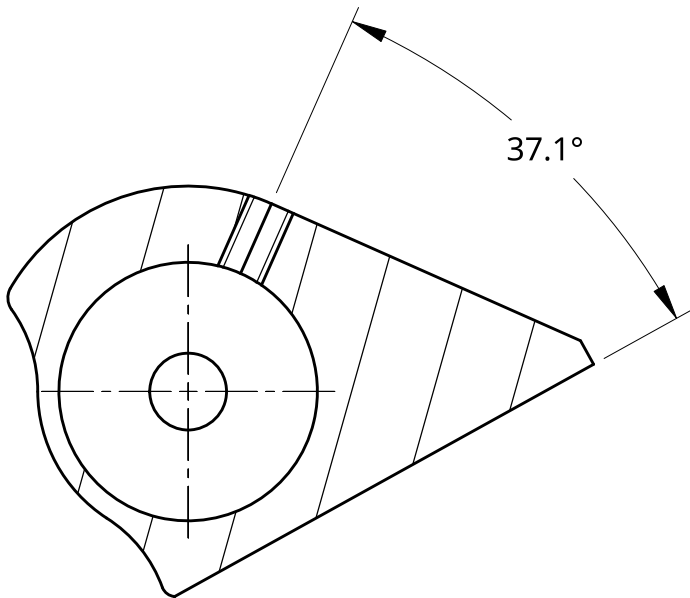
C

B

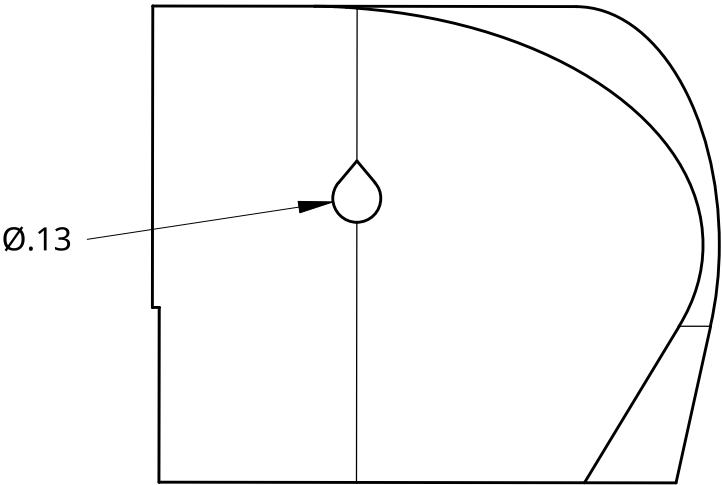
B

A

A



SECTION B - B



Auxiliary C - C

4

3

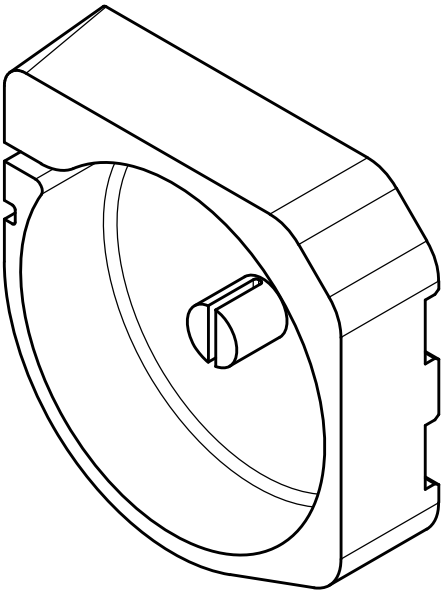
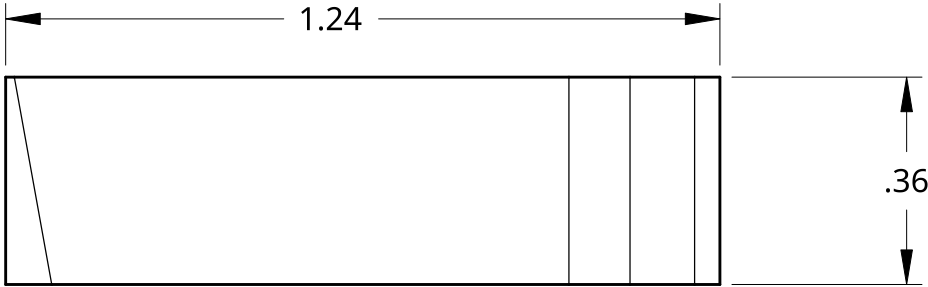
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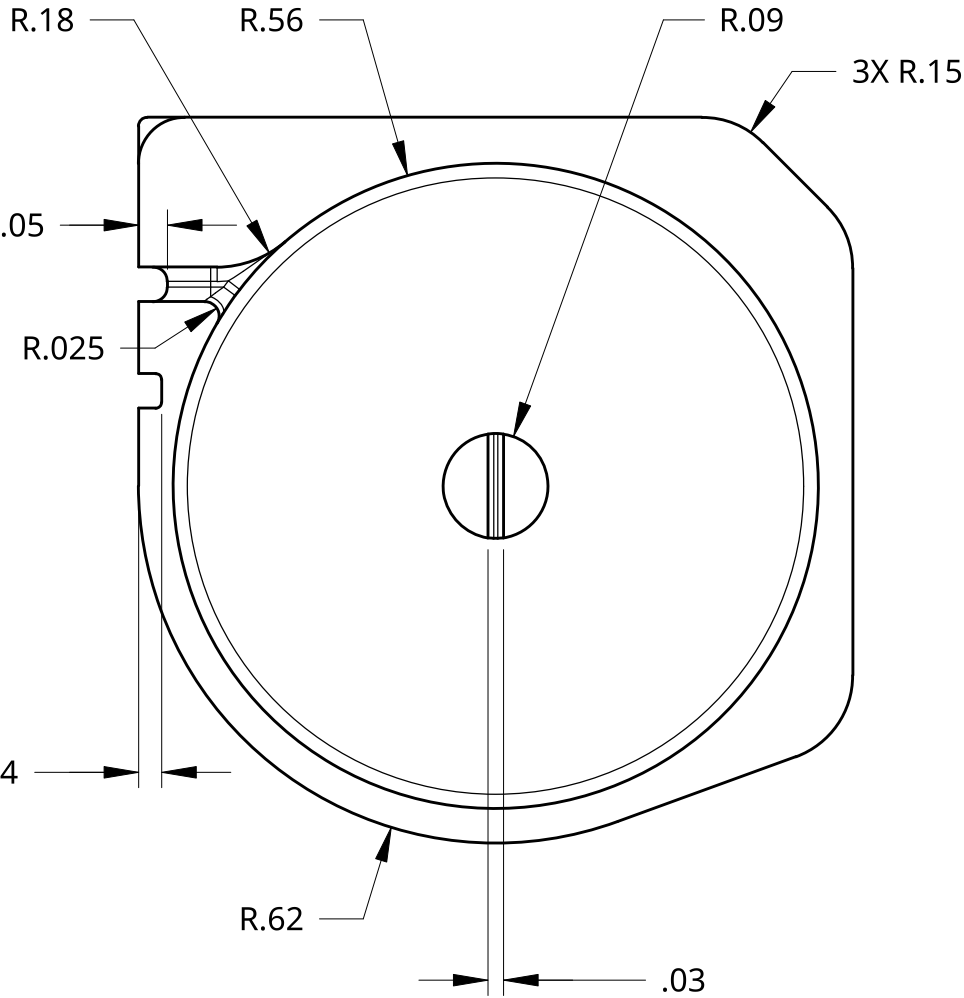
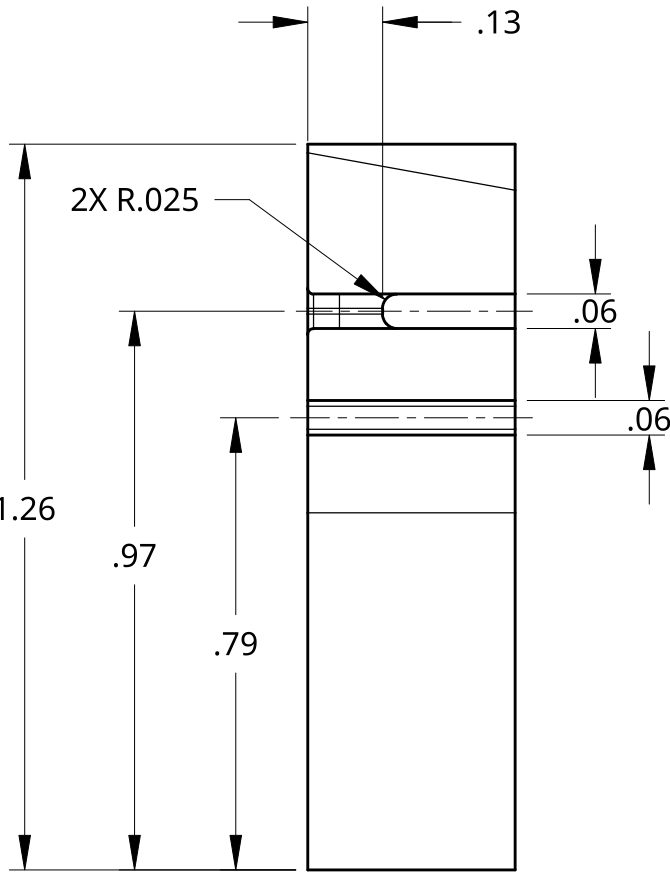
CONFIDENTIAL. DO NOT REPRODUCE. ALL CONTENT OWNED BY SOVEREIGNESS DESIGNS

REVISION TABLE

REV	REVISION DESCRIPTION	DATE
A	INITIAL RELEASE	4 JUL 2025

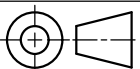


SCALE 2:1



NOTES:

1. THIS PART IS 3D PRINTED. THE SOLID MODEL COMPRISES A COMPLETE DESIGN DOCUMENT. THIS DOCUMENT HAS ADDITIONAL SPECIFICATIONS AND INSPECTION CRITERIA.

UNLESS OTHERWISE SPECIFIED, DIMENSIONS ARE IN INCHES .X = ±.1 .XX = ±.01 .XXX = ±.001 .XXXX = ±.0005 ANGULAR = ± 1° FRACTIONAL = ±1/32	NAME		DATE		TITLE SPOOL CASE							
	DRAWN		JESSICA BOYERS					3 JUL 2025				
DO NOT SCALE DRAWING							SIZE B					
THIRD ANGLE PROJECTION			MATERIAL		FINISH					DWG NO. DO1018		
			POLYMAKER PLA PRO [BLACK]		AS PRINTED							
							SCALE 3:1		SHEET 1 of 2		REV. A	

4

3

2

1

D

D

C

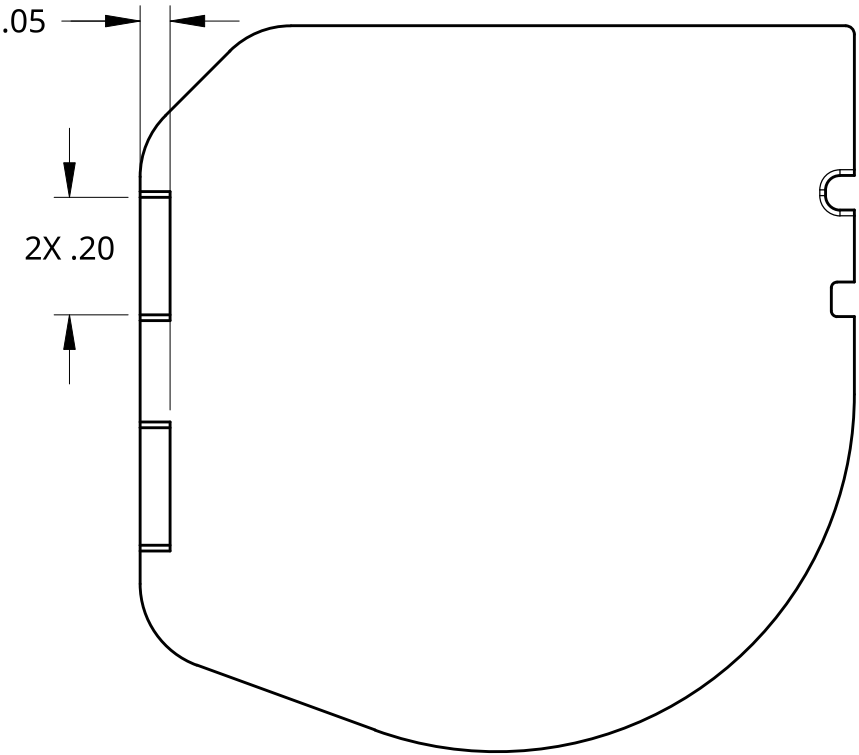
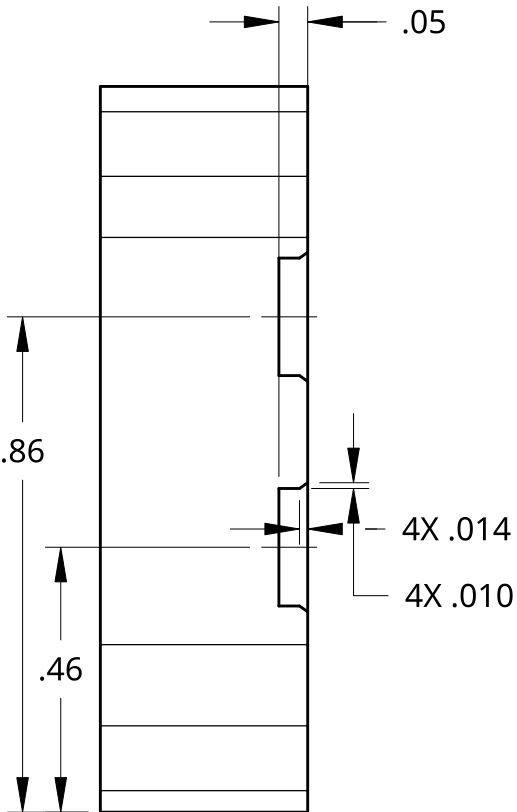
C

B

B

A

A



4

3

2

1